Topic 2
The Brain and the Nervous System

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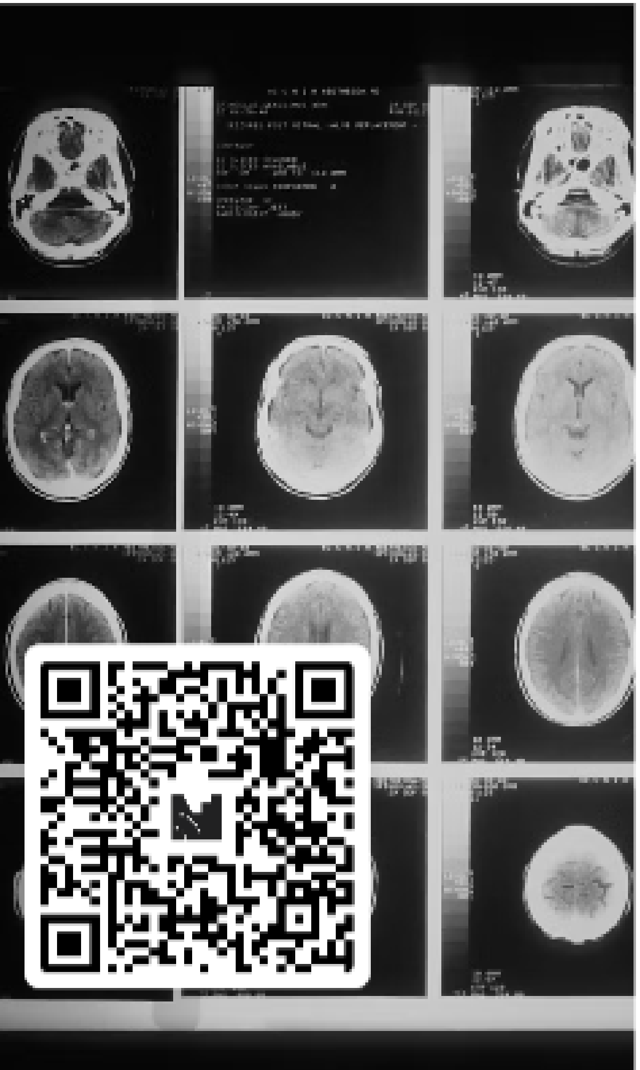
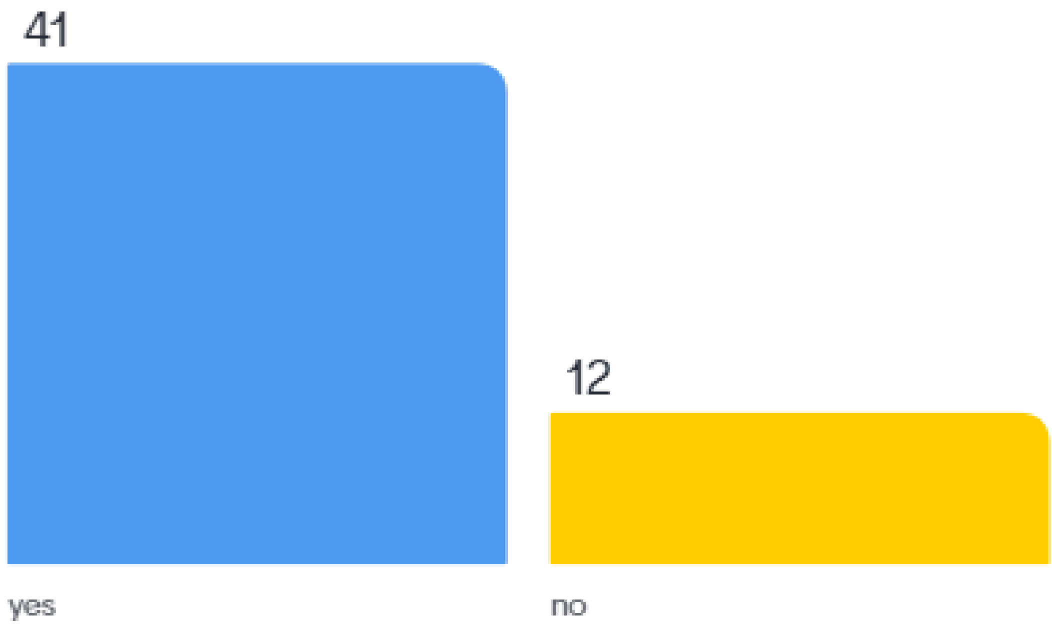


Based on the linked article, would welcome the possibility to predict behavior based on brain scans (e.g. predict future criminal behaviour)?

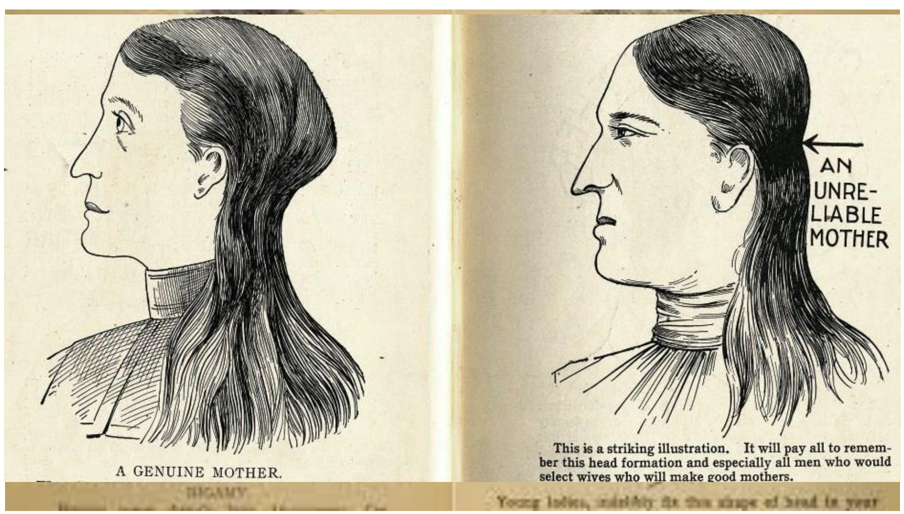


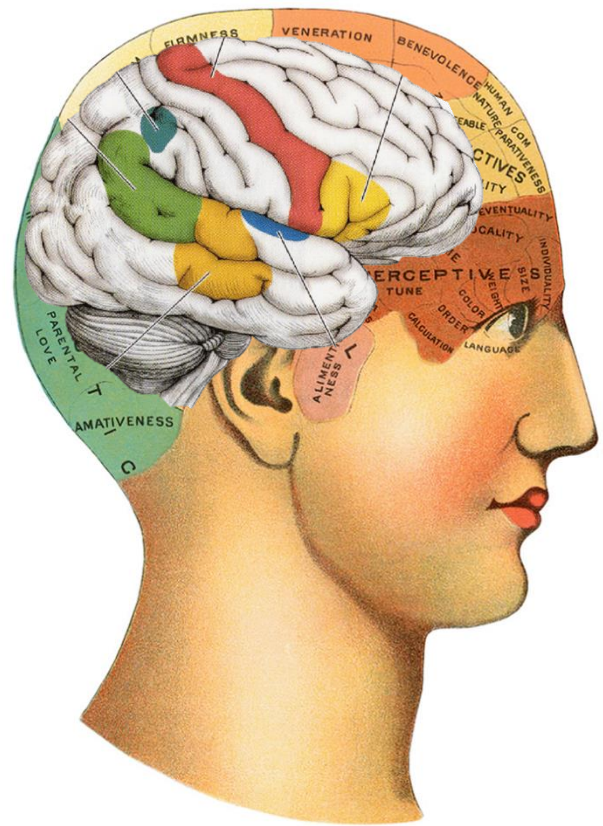
Overview

- Studying the brain
- Building blocks of the nervous system
- Communication among neurons
- Communication of the brain with the body
- The brain

Studying the brain

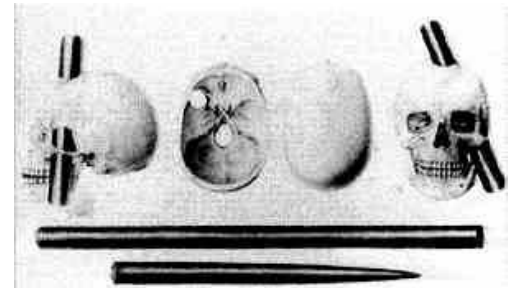
- 19th century -> Phrenology
  - Bumps on the skull were interpreted in terms of personality traits.

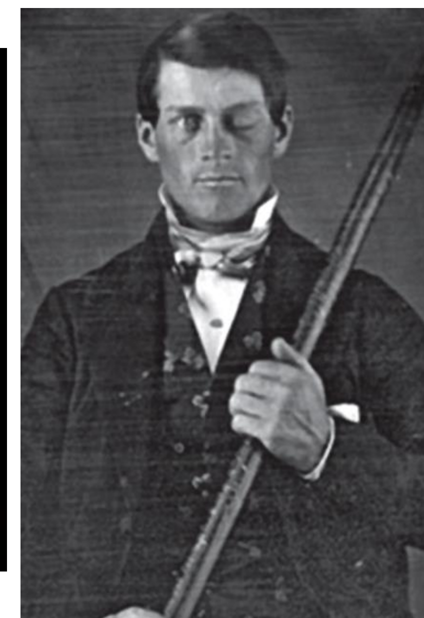




The Case of Phineas Gage

https://www.youtube.com/watch?v=yXbAMHzYGJ0&t=2s&ab_channel=HarvardUniversity





Studying the brain

- Methods for studying the brain
  - Clinical observation of patients with brain damage
  - Experimental techniques
    - Invasive: animal studies (e.g., lesions, single-cell recordings)

Studying the brain

Figure 3.4
David Hubel (left) and Torsten Wiesel (right), who won the 1981 Nobel prize in physiology and medicine for their research on the physiology of vision. The results of their research established the groundwork for much of our knowledge of visual physiology.

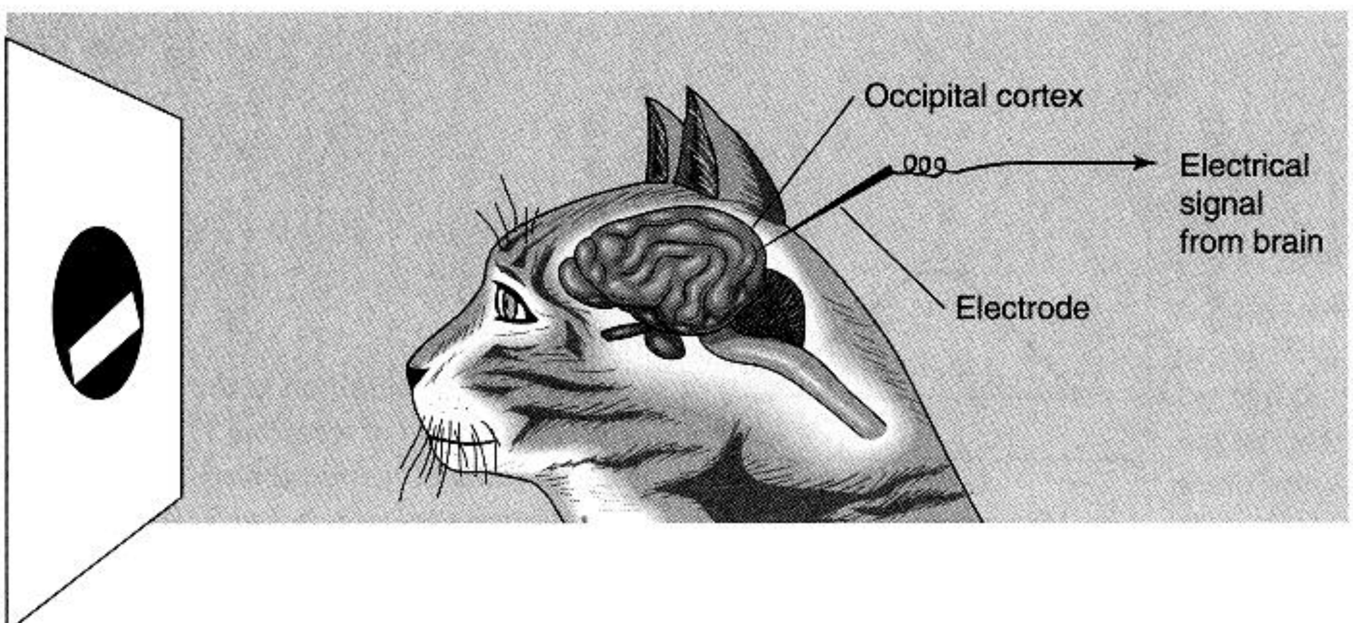


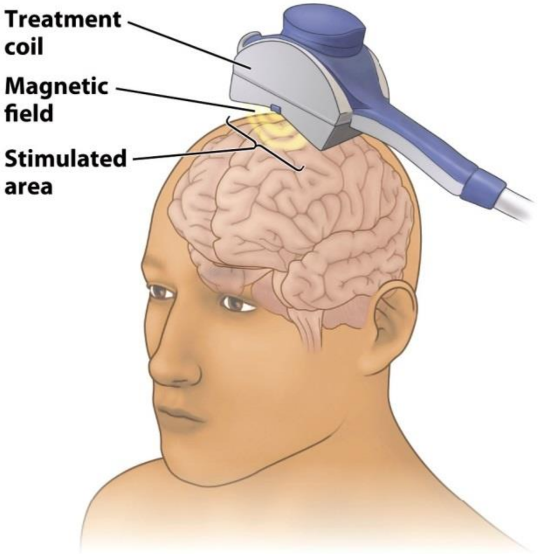
Figure 2.33
Recording electrical signals from a neuron in the visual cortex of an anaesthetized cat. The bar-shaped stimulus on the screen causes a nerve cell in the cortex to fire, and a recording electrode picks up the signals generated by this nerve cell. In an actual experiment, the cat is anesthetized and its head is held in place for accurate positioning.

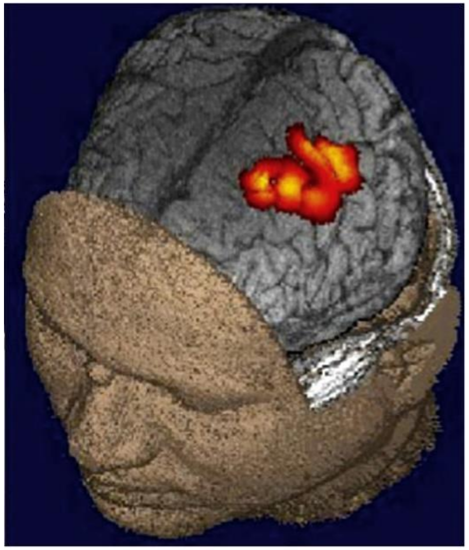
Studying the brain

- Methods for studying the brain
  - Clinical observation of patients with brain damage
  - Experimental techniques
    - Invasive: animal studies (e.g., lesions, single-cell recordings)
    - TMS (Transcranial Magnetic Stimulation)

Studying the brain

TMS





Studying the brain

- Methods for studying the brain
  - Clinical observation of patients with brain damage
  - Experimental techniques
    - Invasive: animal studies (e.g., lesions, single-cell recordings)
    - TMS (Transcranial Magnetic Stimulation)
  - Other techniques
    - Electrophysiology
      - EEG (ERP)

Studying the brain

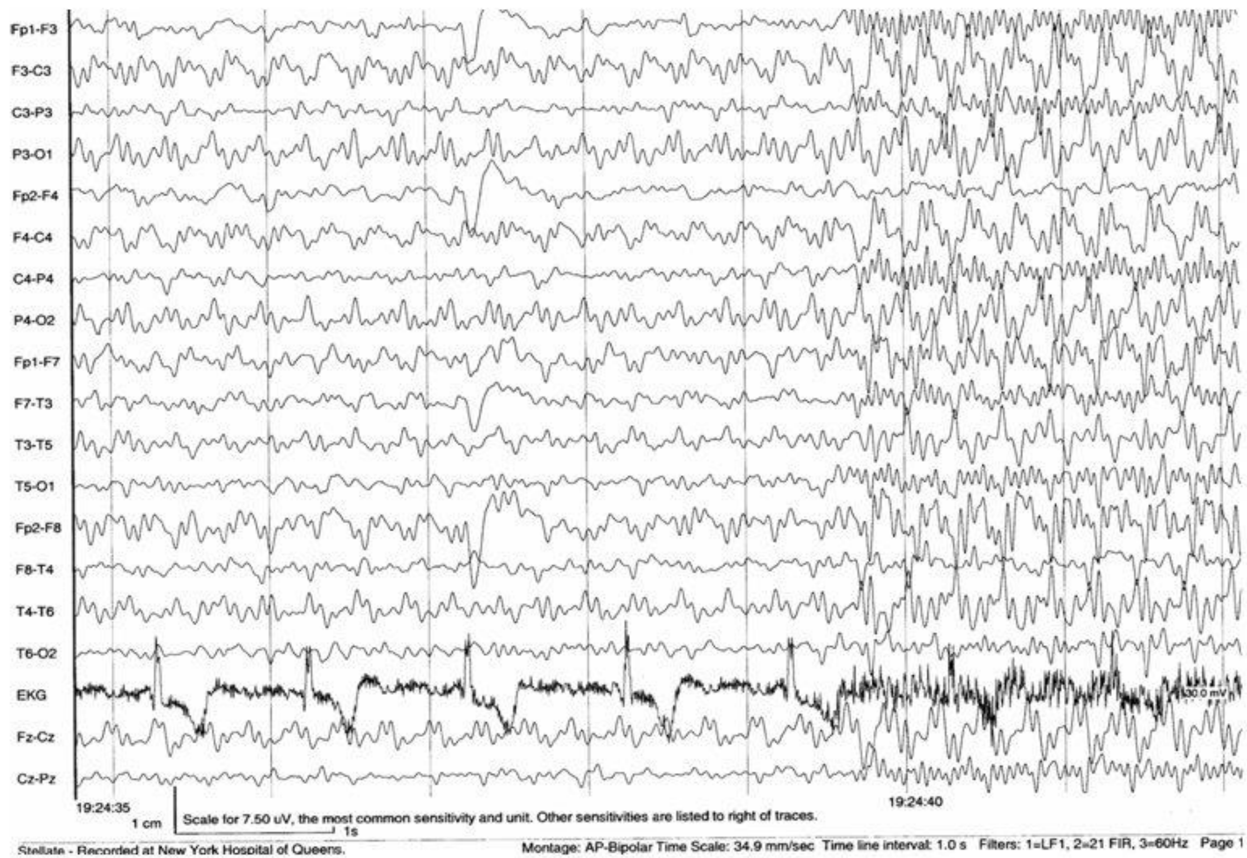
EEG

1 – 256 channels

common is 32, 64, 128

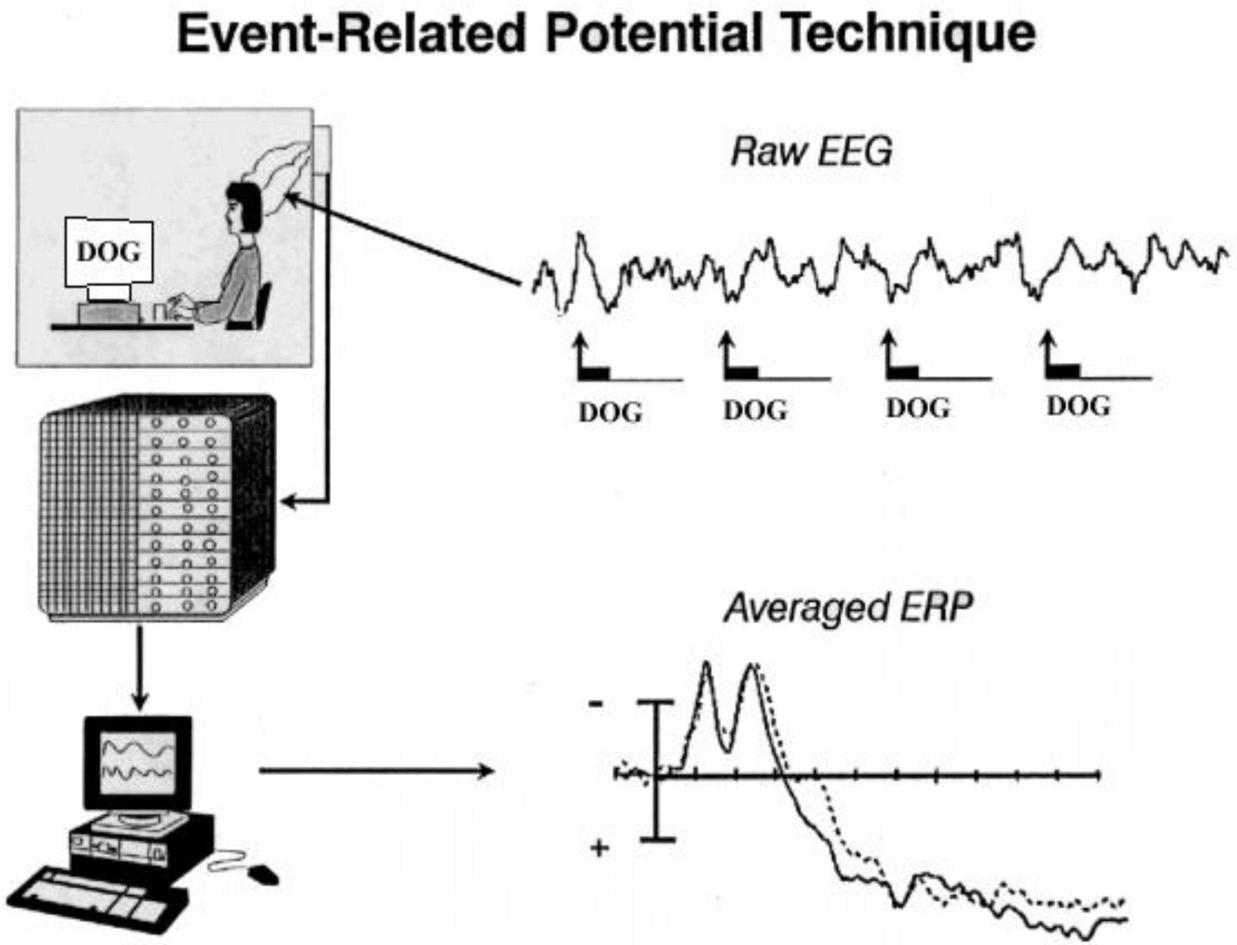
Studying the brain

EEG



Studying the brain

ERP

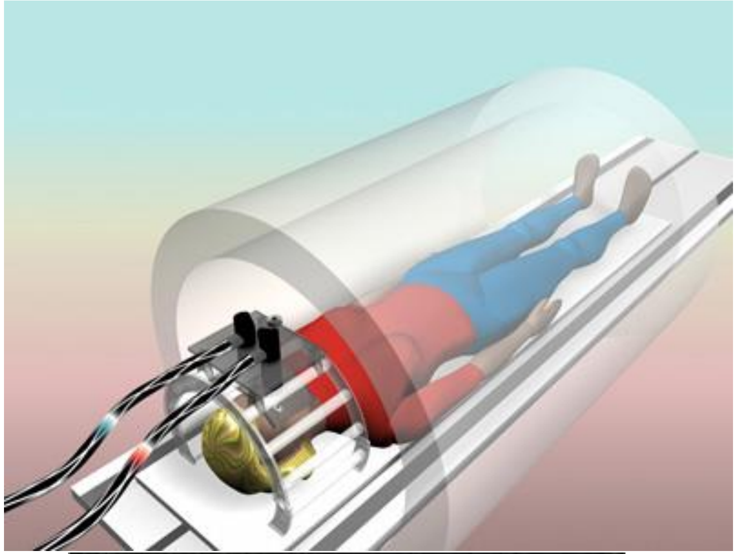


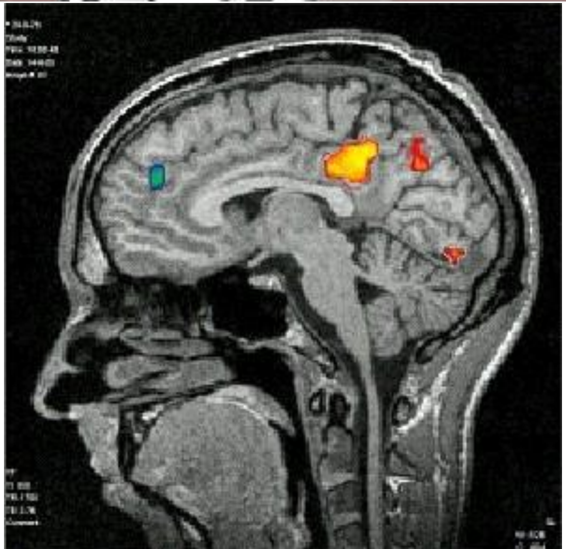
Studying the brain

- Methods for studying the brain
  - Clinical observation of patients with brain damage
  - Experimental techniques
    - Invasive: animal studies (e.g., lesions, single-cell recordings)
    - TMS (Transcranial Magnetic Stimulation)
  - Other techniques
    - Electrophysiology
      - EEG (ERP)
    - Brain imaging
      - (f)MRI scan
      - PET scan

Studying the brain

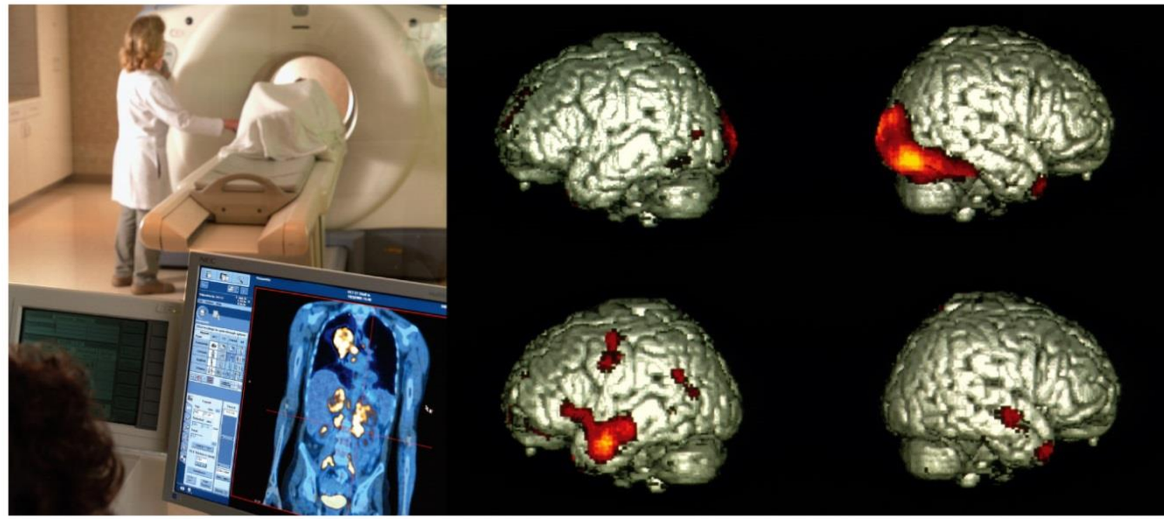
fMRI

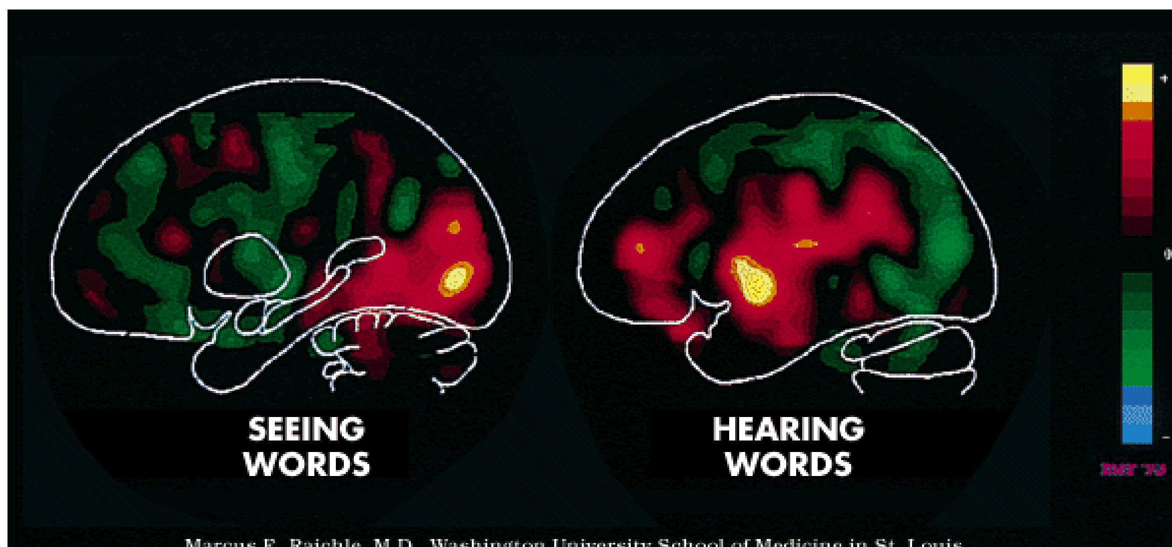




Studying the brain

PET

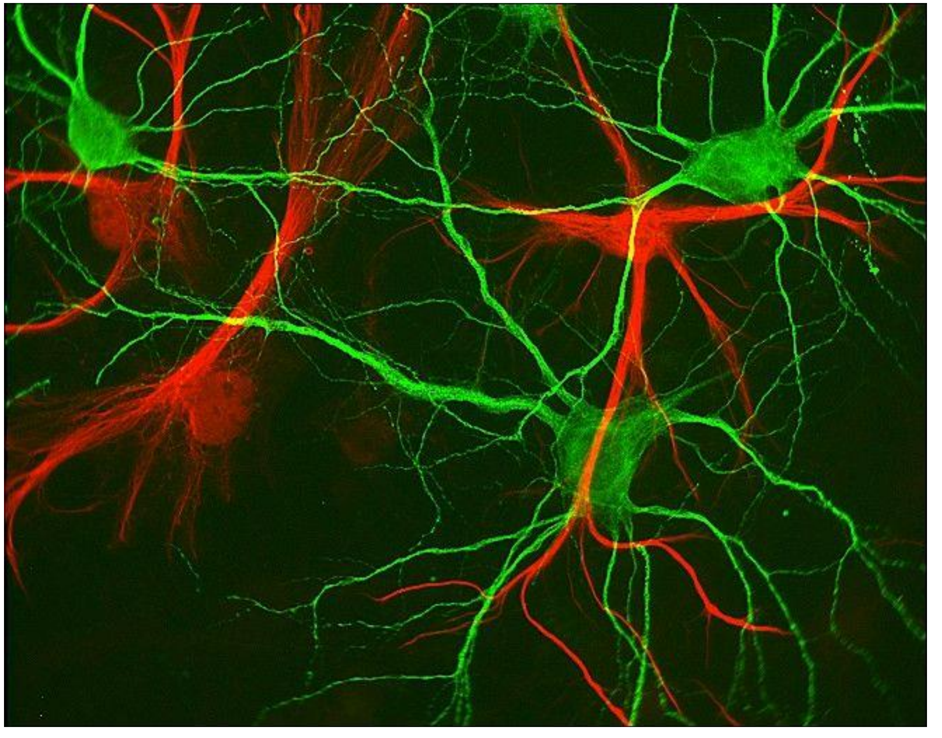




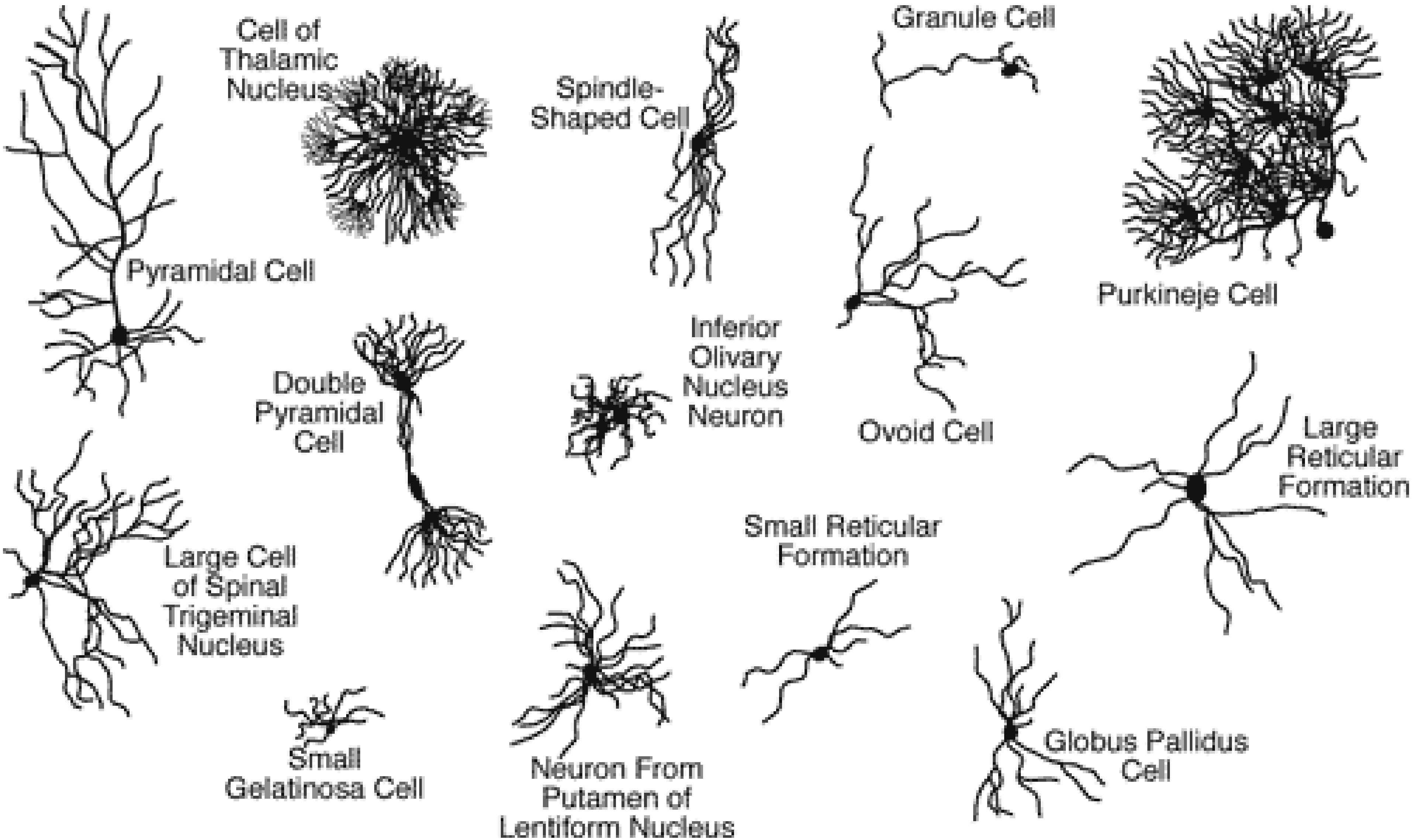
- Genes x environment

Building blocks of the nervous system

- The nervous system is made up of two basic kinds of cells
  - Glia
  - Neurons

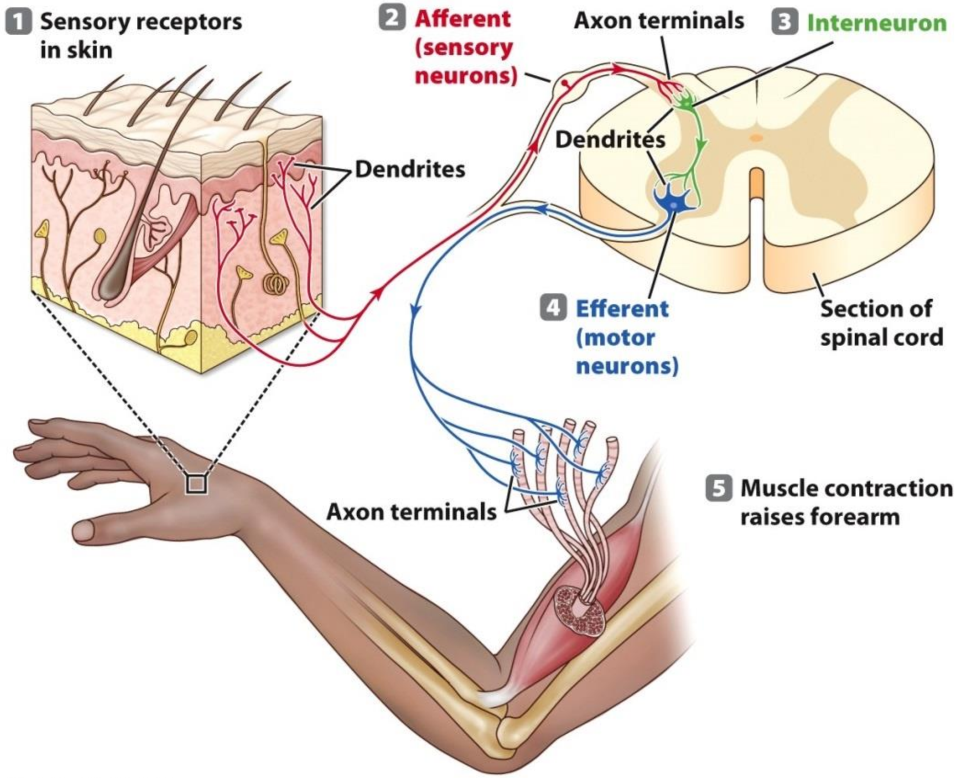


Building blocks of the nervous system

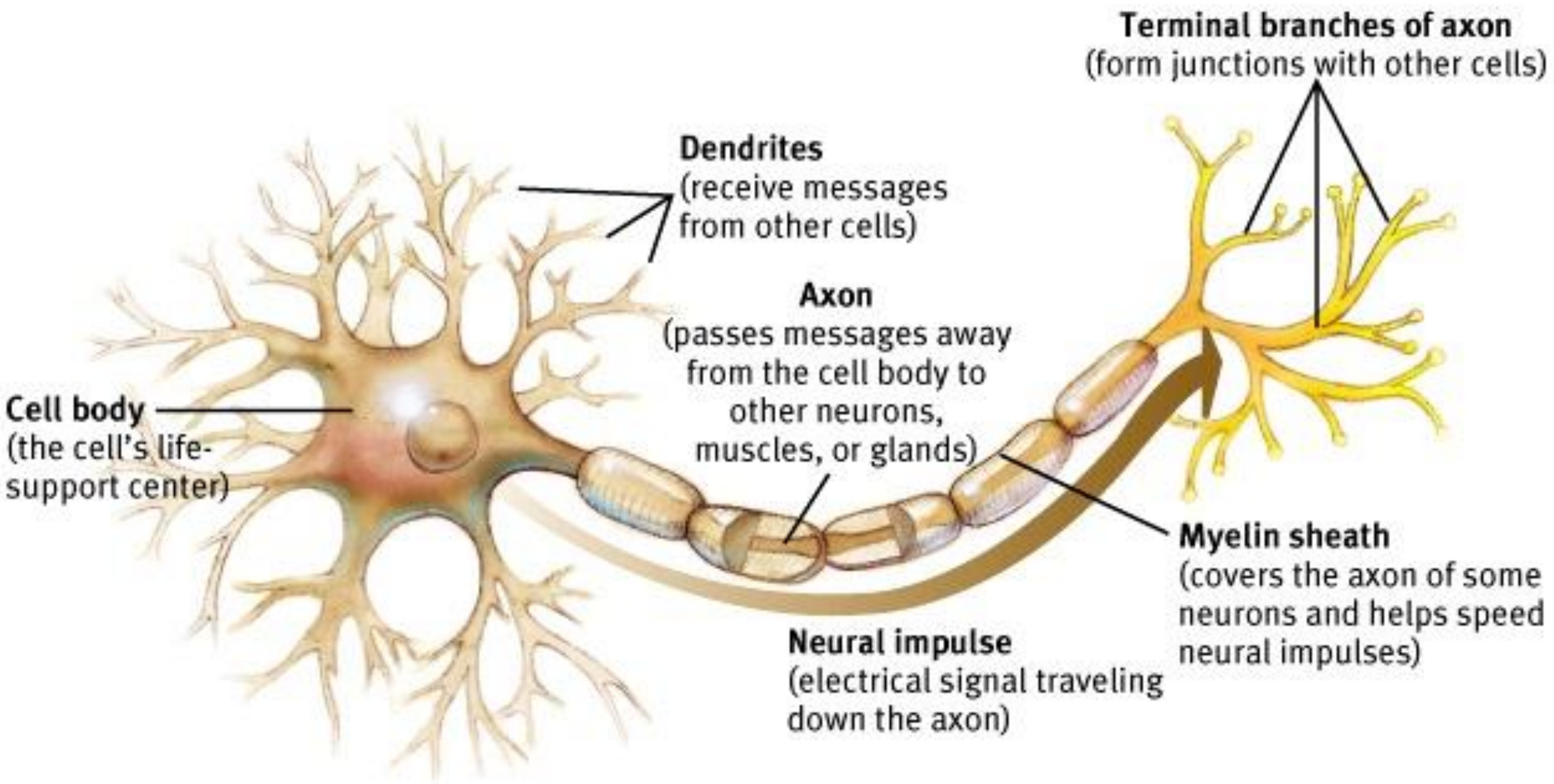


Building blocks of the nervous system

- Different types of neurons
  - Sensory receptors
  - Sensory (afferent) neurons
  - Motor (efferent) neurons
  - Interneurons

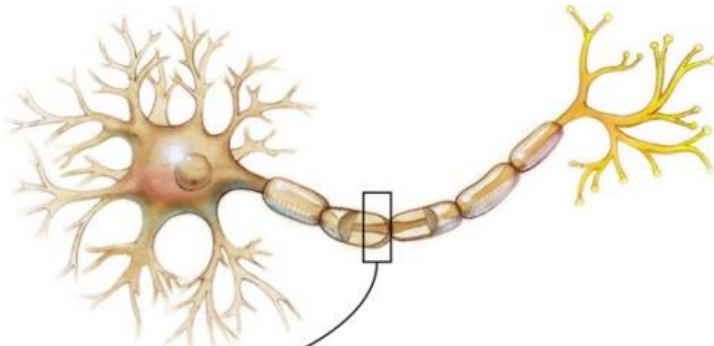


Building blocks of the nervous system:
The neuron

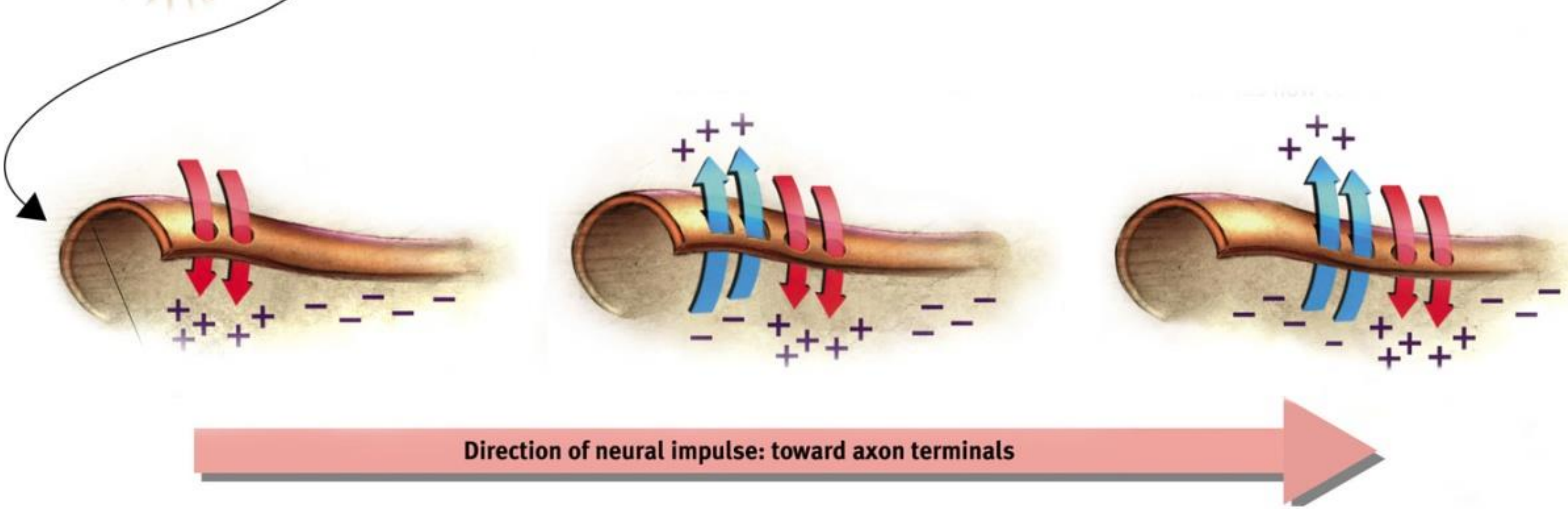


Communication among neurons

Action Potential: is a **neural impulse**, a brief electrical charge that travels down an axon.

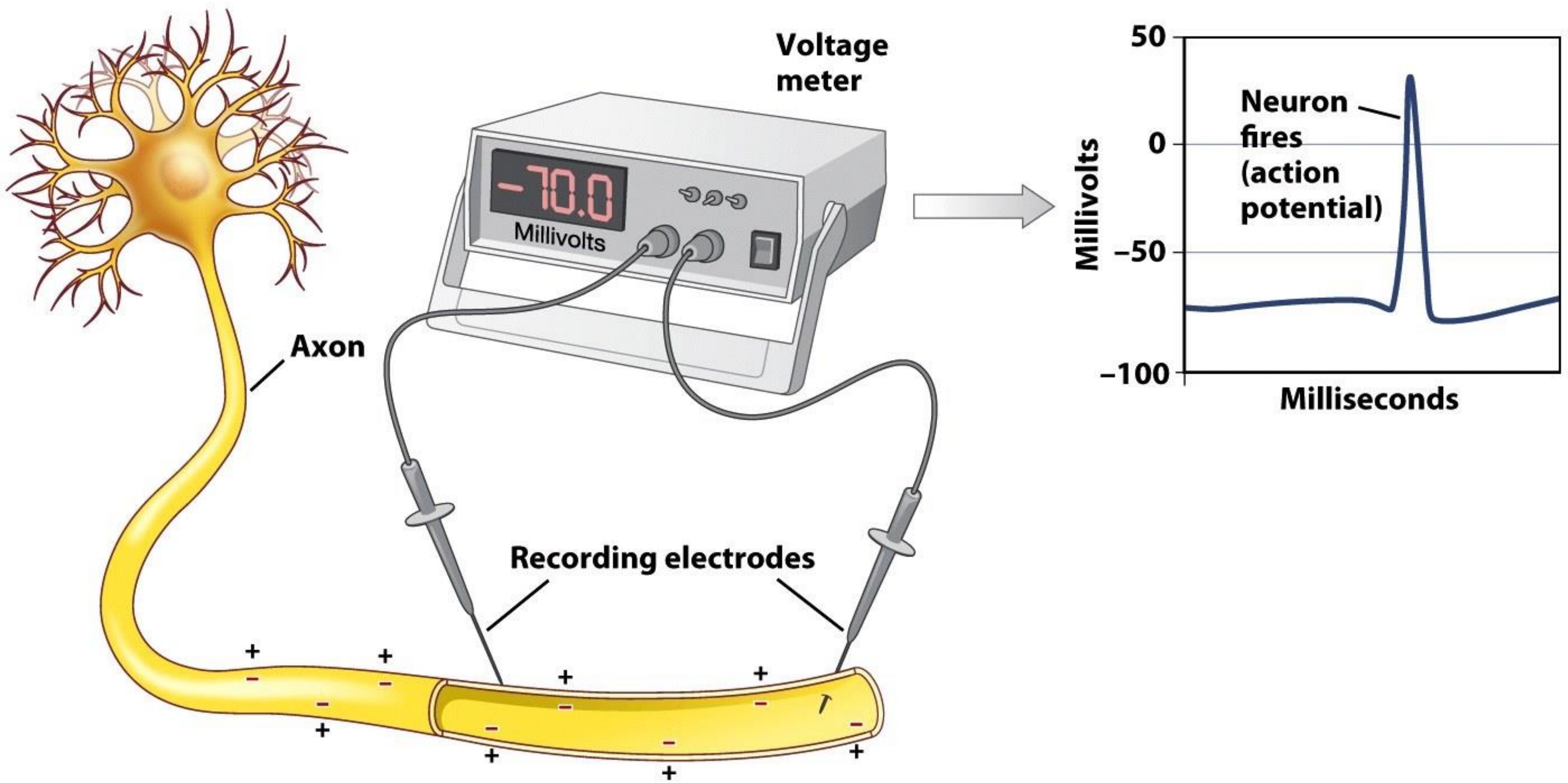


It is very fast (3 to 320 km per h)

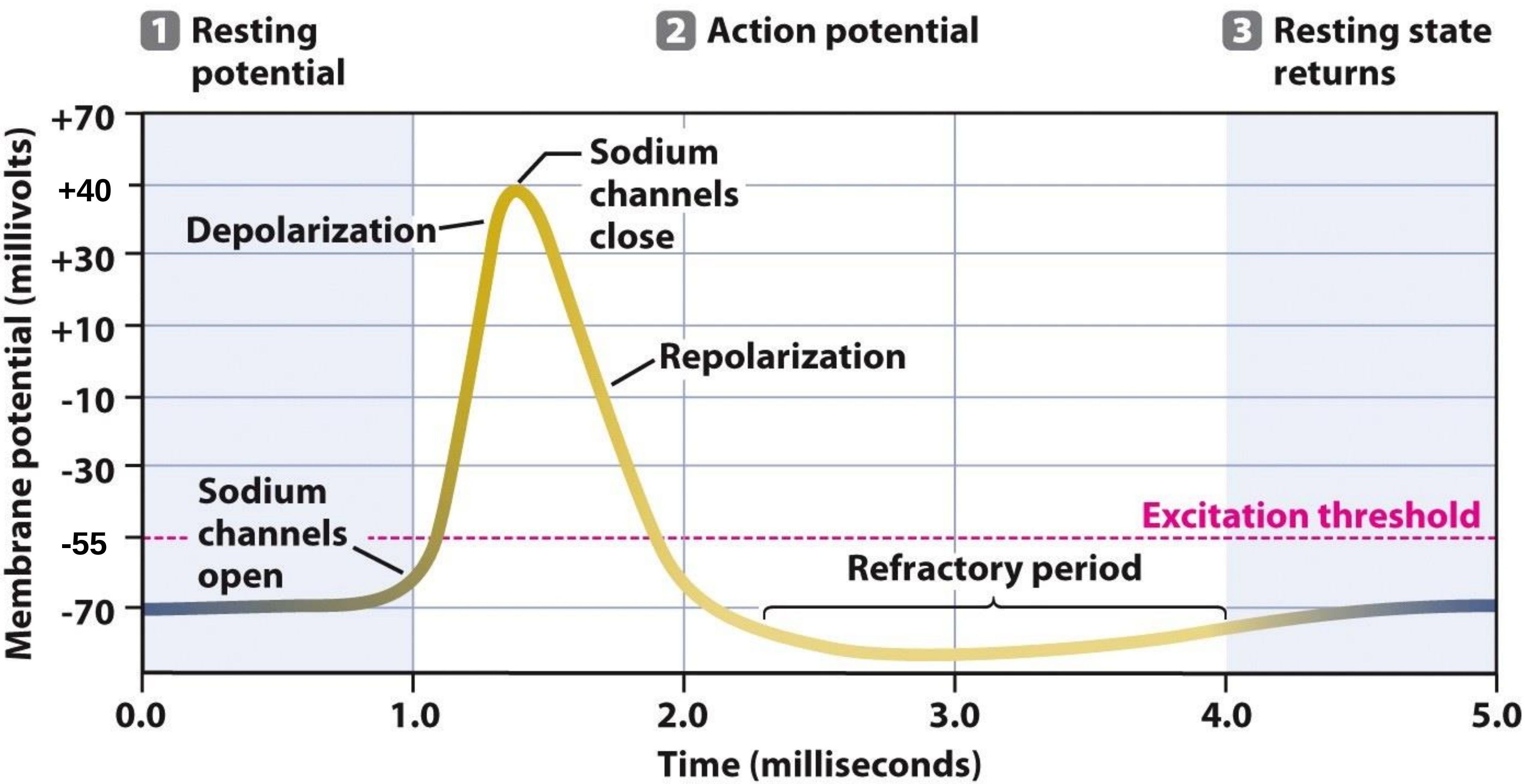


One direction

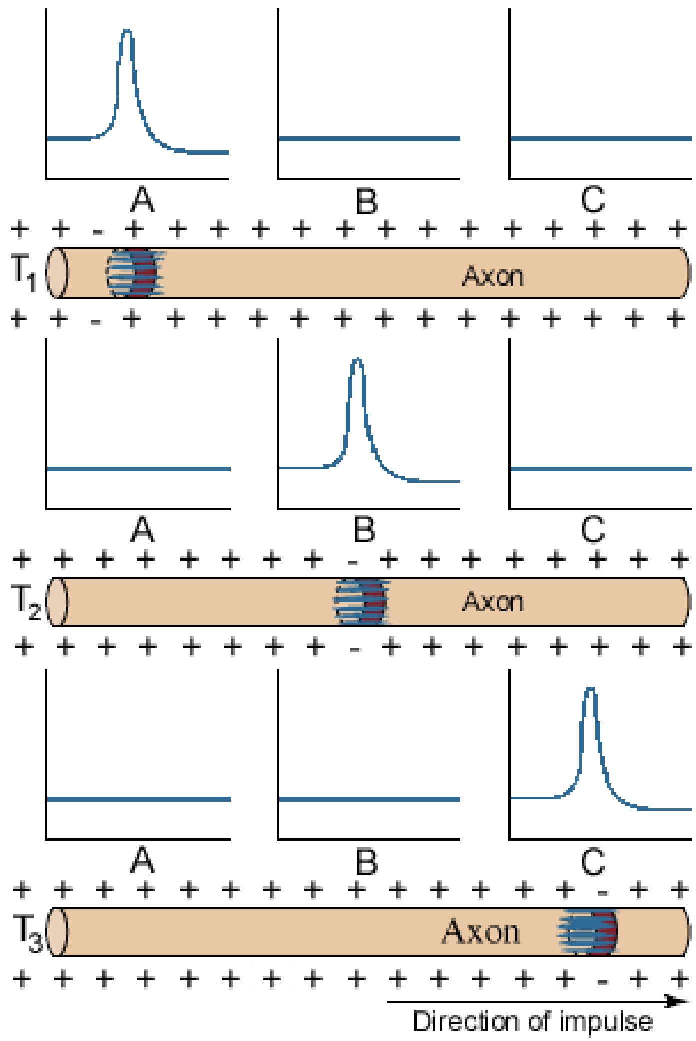
Communication among neurons



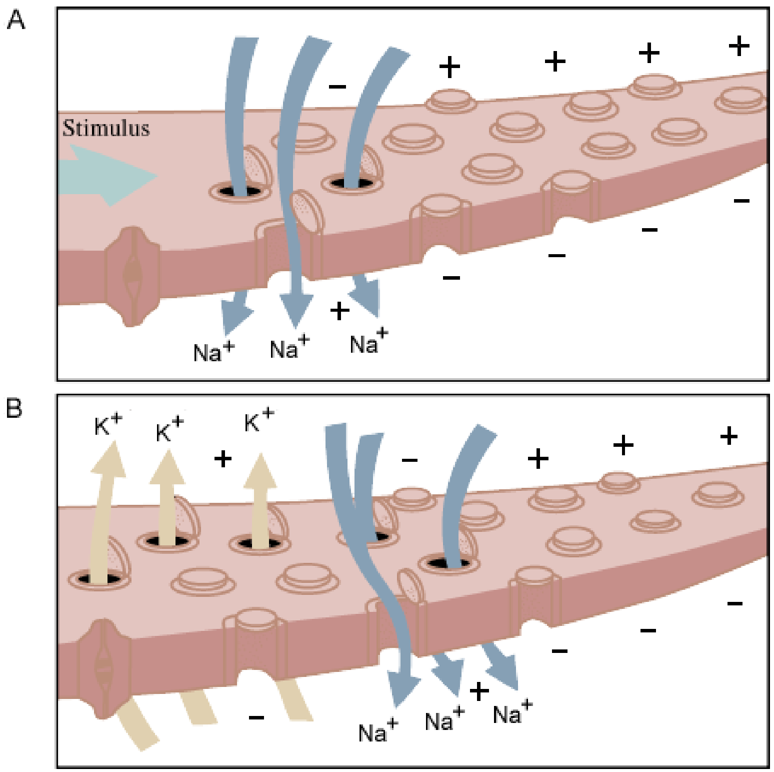
Communication among neurons

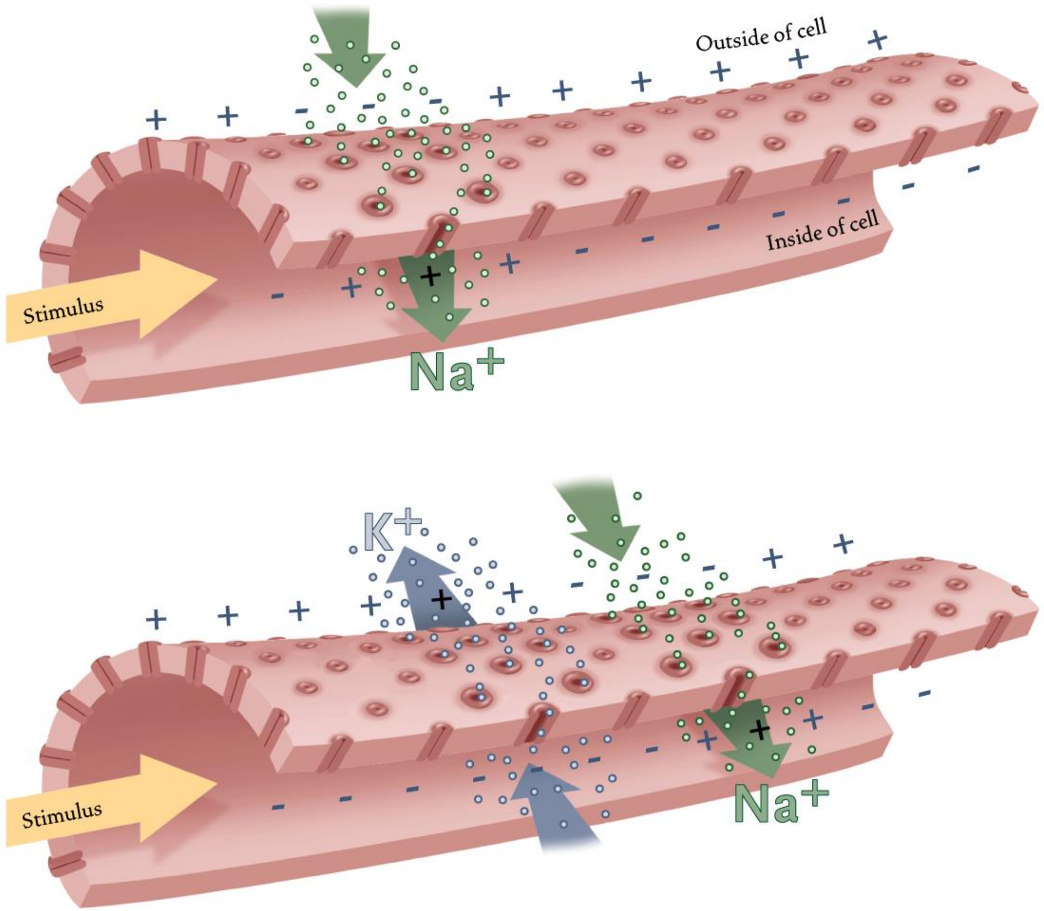


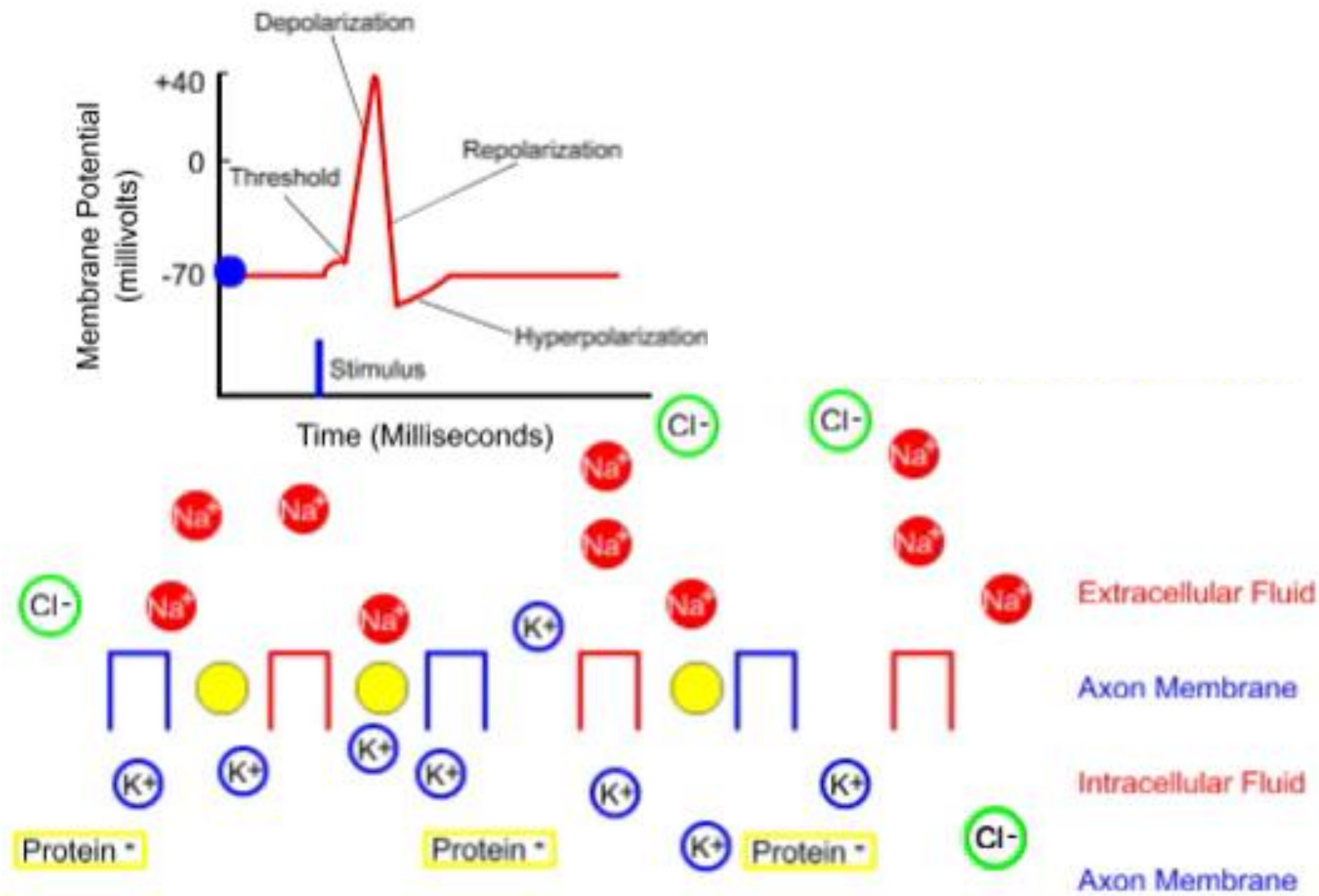
Communication among neurons



Communication among neurons





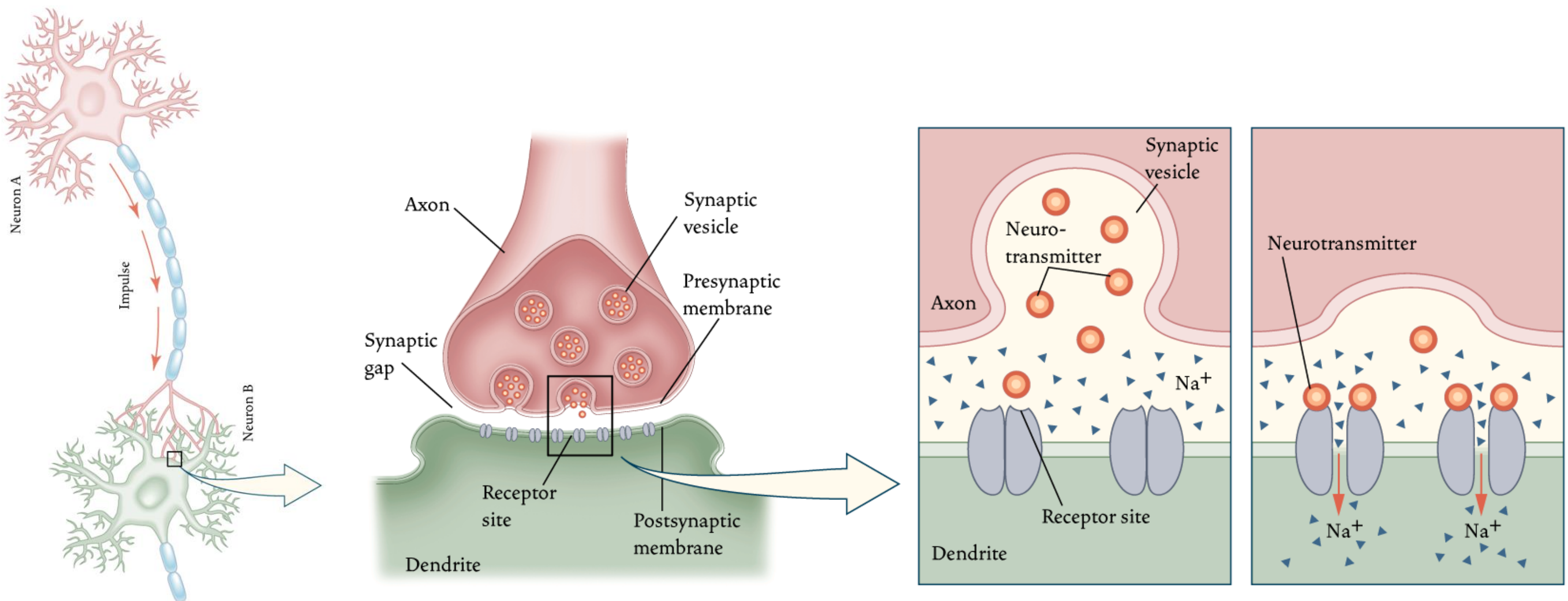

Depolarization
+40
0
-70
Membrane Potential (millivolts)
Threshold
Repolarization
Hyperpolarization
Stimulus
Time (Milliseconds)
Cl-
Na+
K+
Extracellular Fluid
Axon Membrane
Intracellular Fluid
Protein -
Axon Membrane

Communication among neurons

- Neurons either fire or do not fire
  - All-or-none law
    - Intensity variations by
      - variations in the number of neurons firing (number).
      - variations in firing rate (frequency).
- Neurons interact
  - via synapses.
  - through chemical substances.

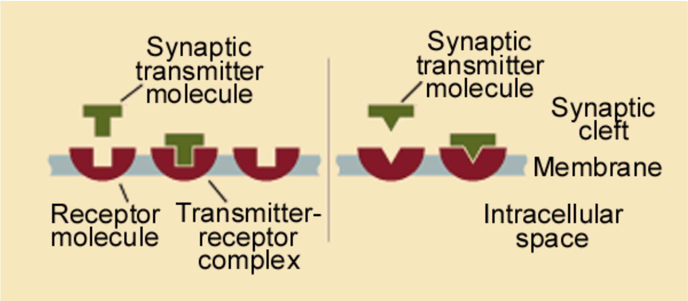
Communication among neurons

- Synapse
  - the place where a signal passes from one nerve cell to another

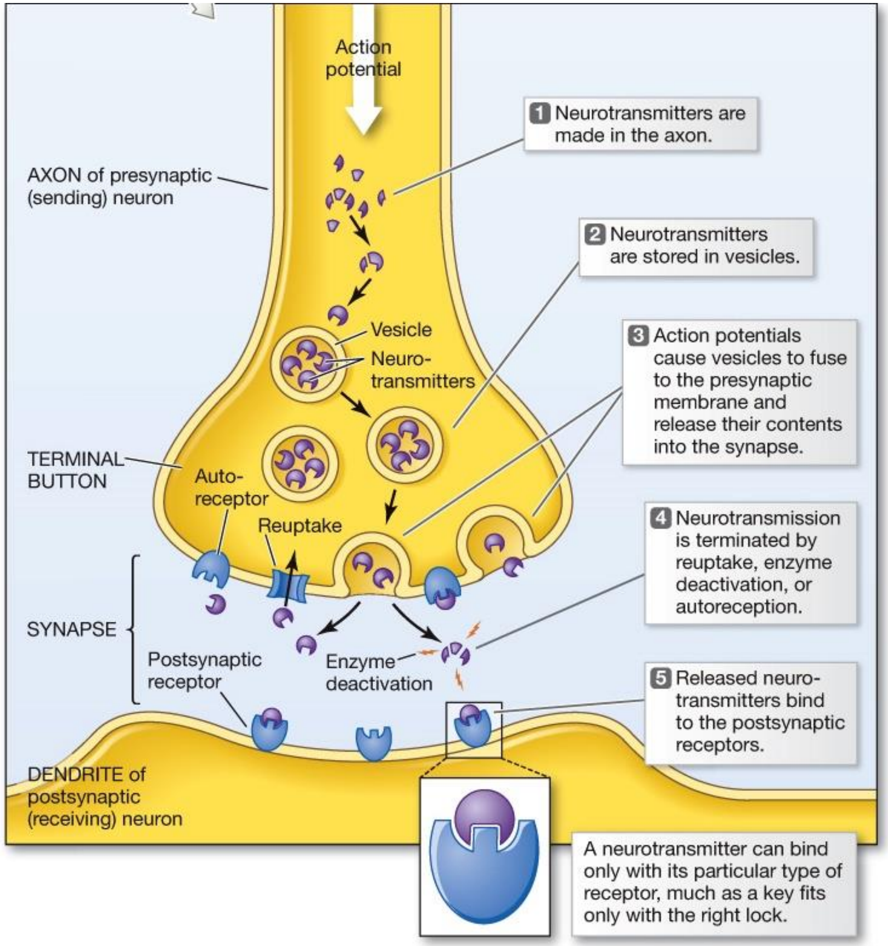


Communication among neurons

- Neurotransmitters
  - Chemical substances that transmit signals from one neuron to another
    - Lock-and-key Model

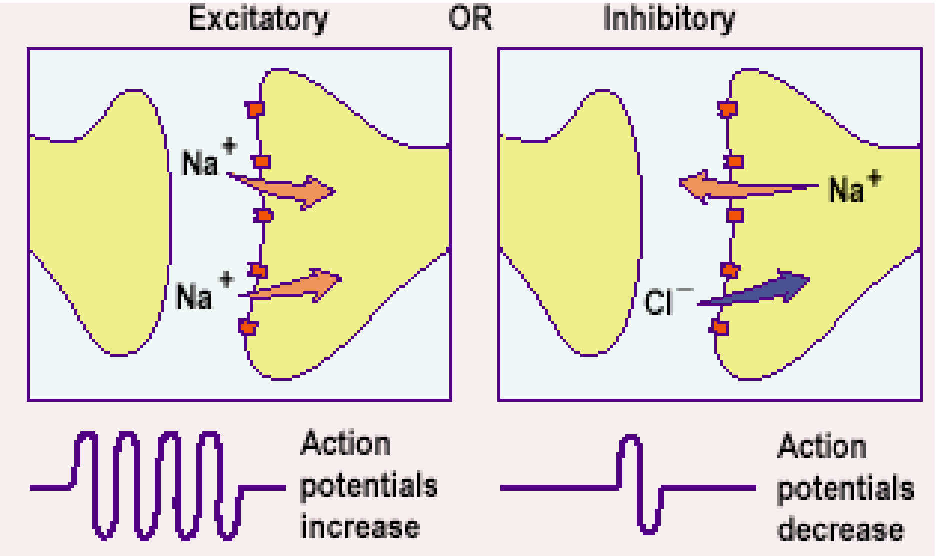


    - Effect is terminated by
      - autoreceptors
      - synaptic reuptake
      - Enzyme deactivation



Communication among neurons

- Neurotransmitters
  - Chemical substances that transmit signals from one neuron to another
    - The binding of a neurotransmitter with a receptor produces an excitatory or inhibitory signal.



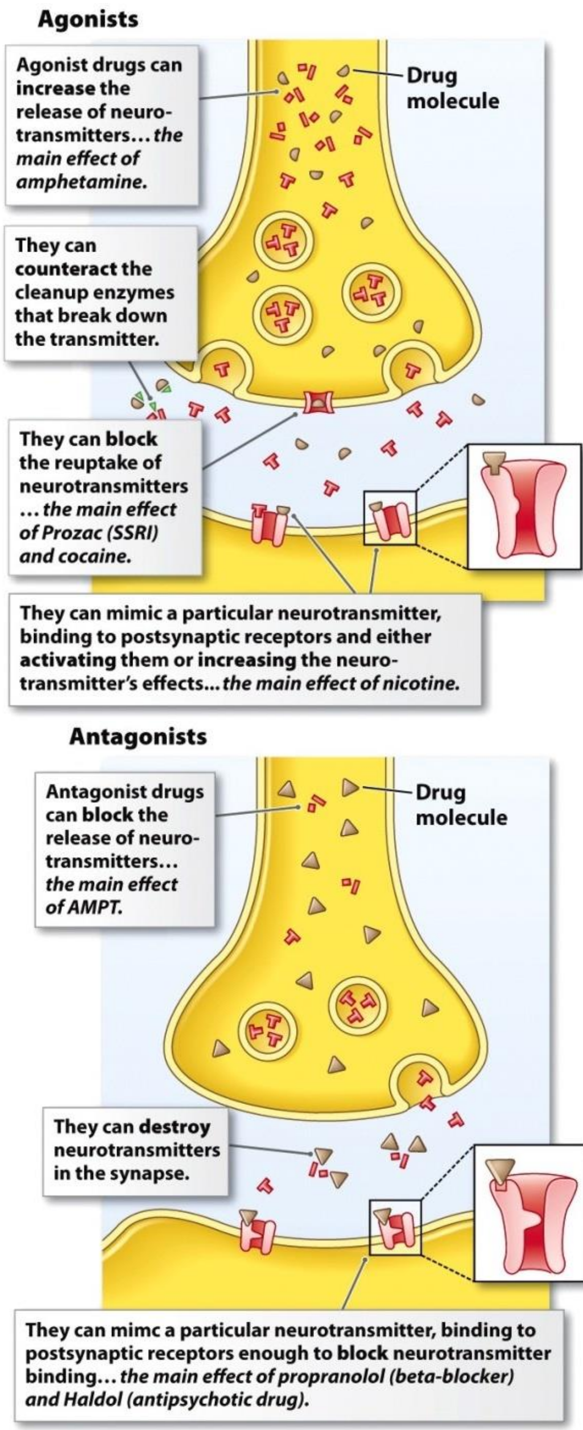
Communication among neurons

- Neurotransmitters

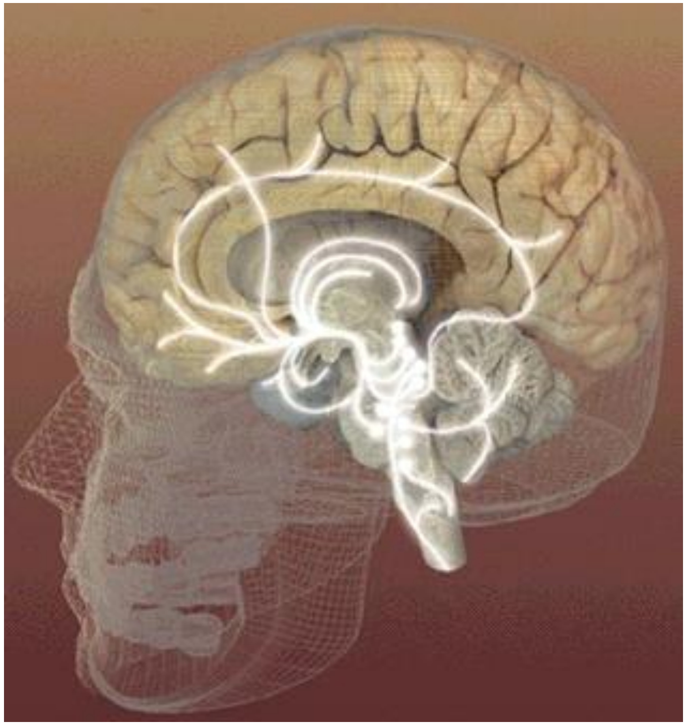
NEUROTRANSMITTER	PSYCHOLOGICAL FUNCTIONS
Acetylcholine	Motor control over muscles
	Learning, memory, sleeping, and dreaming
Epinephrine	Energy
Norepinephrine	Arousal, vigilance, and attention
Serotonin	Emotional states and impulsiveness
	Dreaming
Dopamine	Reward and motivation
	Motor control over voluntary movement
GABA (gamma-aminobutyric acid)	Inhibition of action potentials
	Anxiety reduction
Glutamate	Enhancement of action potentials
	Learning and memory
Endorphins	Pain reduction
	Reward

Communication among neurons

- Drugs
  - Agonists
    - increase of precursor
    - counteracting the cleanup enzymes
    - blocking the re-uptake
    - mimicking the neurotransmitter's action
  - Antagonists
    - decrease precursor (or neurotransmitter)
    - increase effectiveness cleanup enzymes
    - enhance the re-uptake
    - blocking of receptors



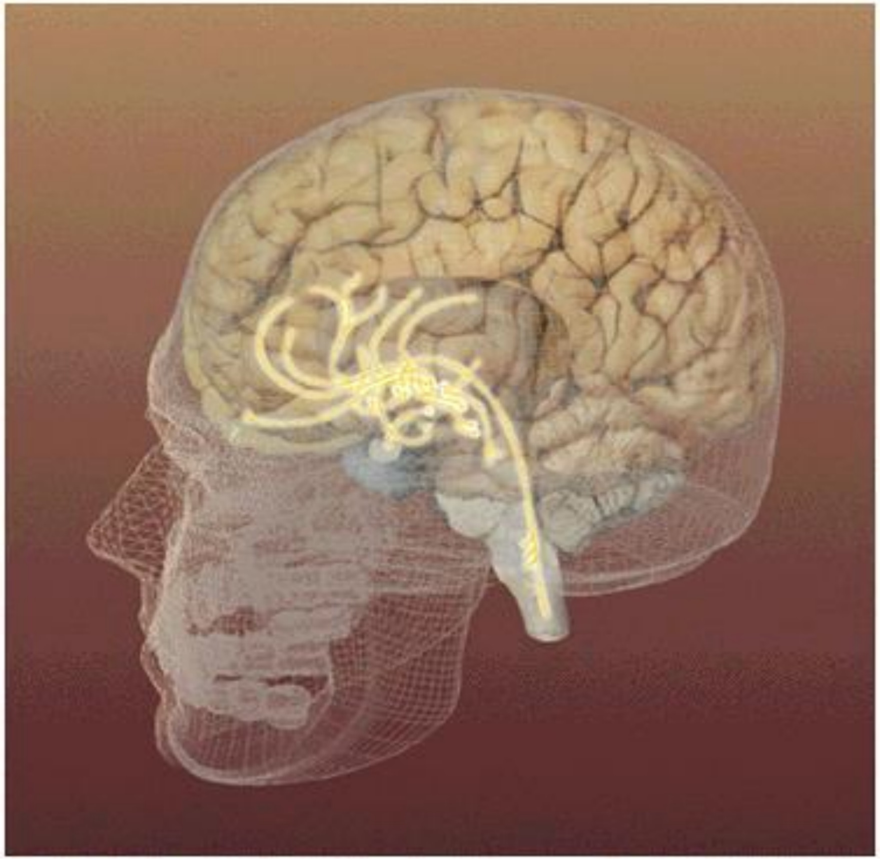
How Neurotransmitters Influence Us



Serotonin pathways are involved with mood regulation.

How Neurotransmitters Influence Us

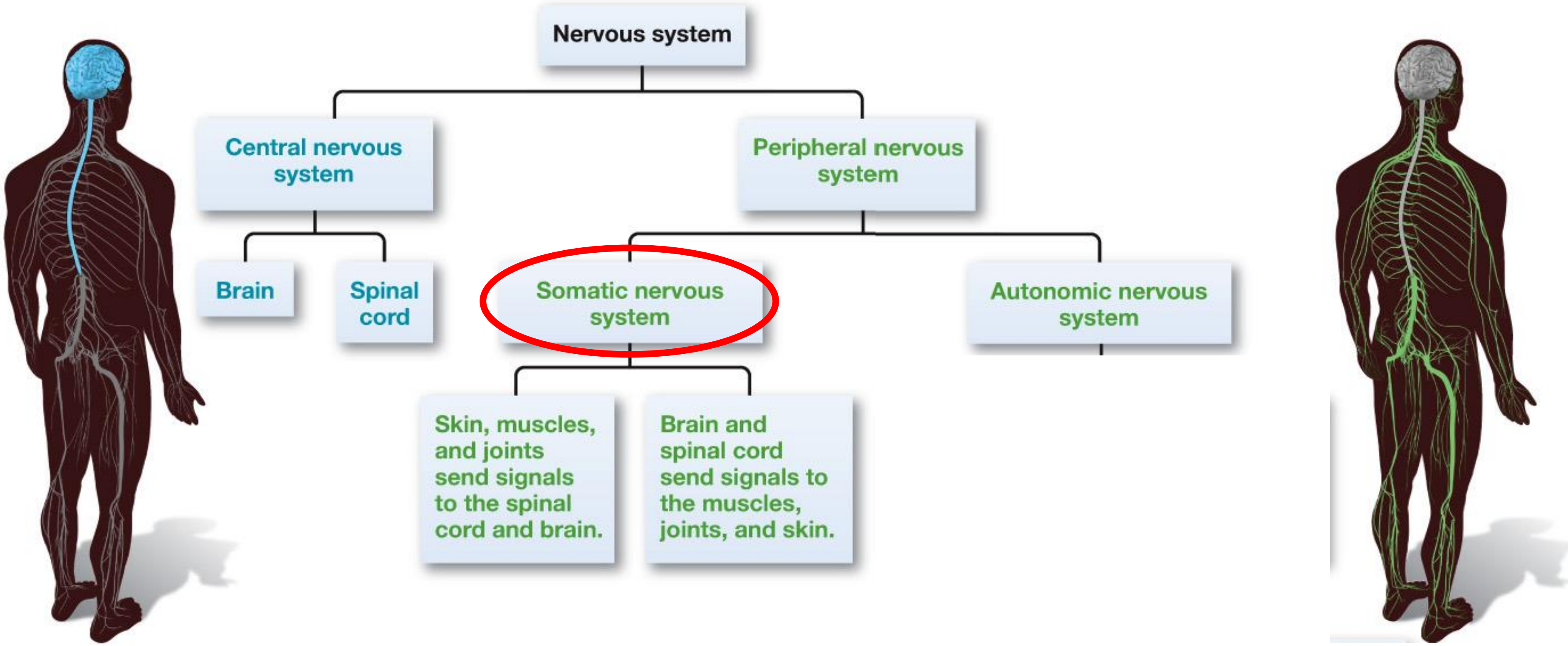
Dopamine pathways are involved with diseases such as schizophrenia and Parkinson's disease.



How Neurotransmitters Influence Us

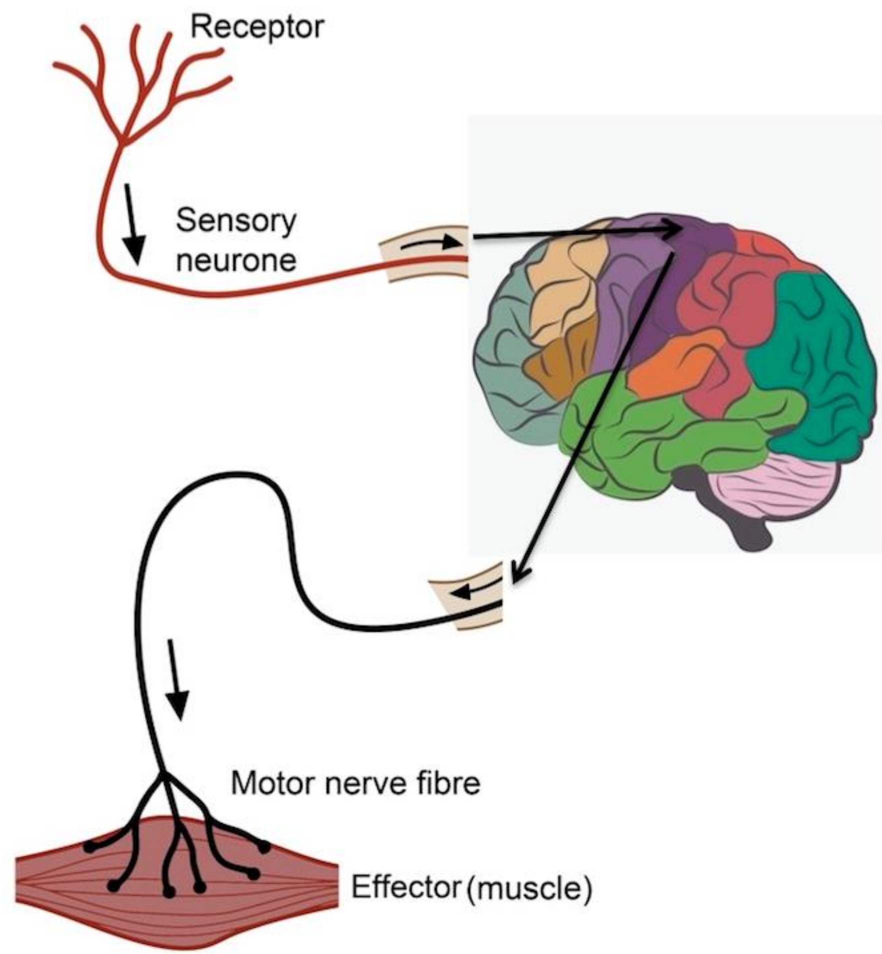
Communication of the brain with the body

Communication of the brain with the body

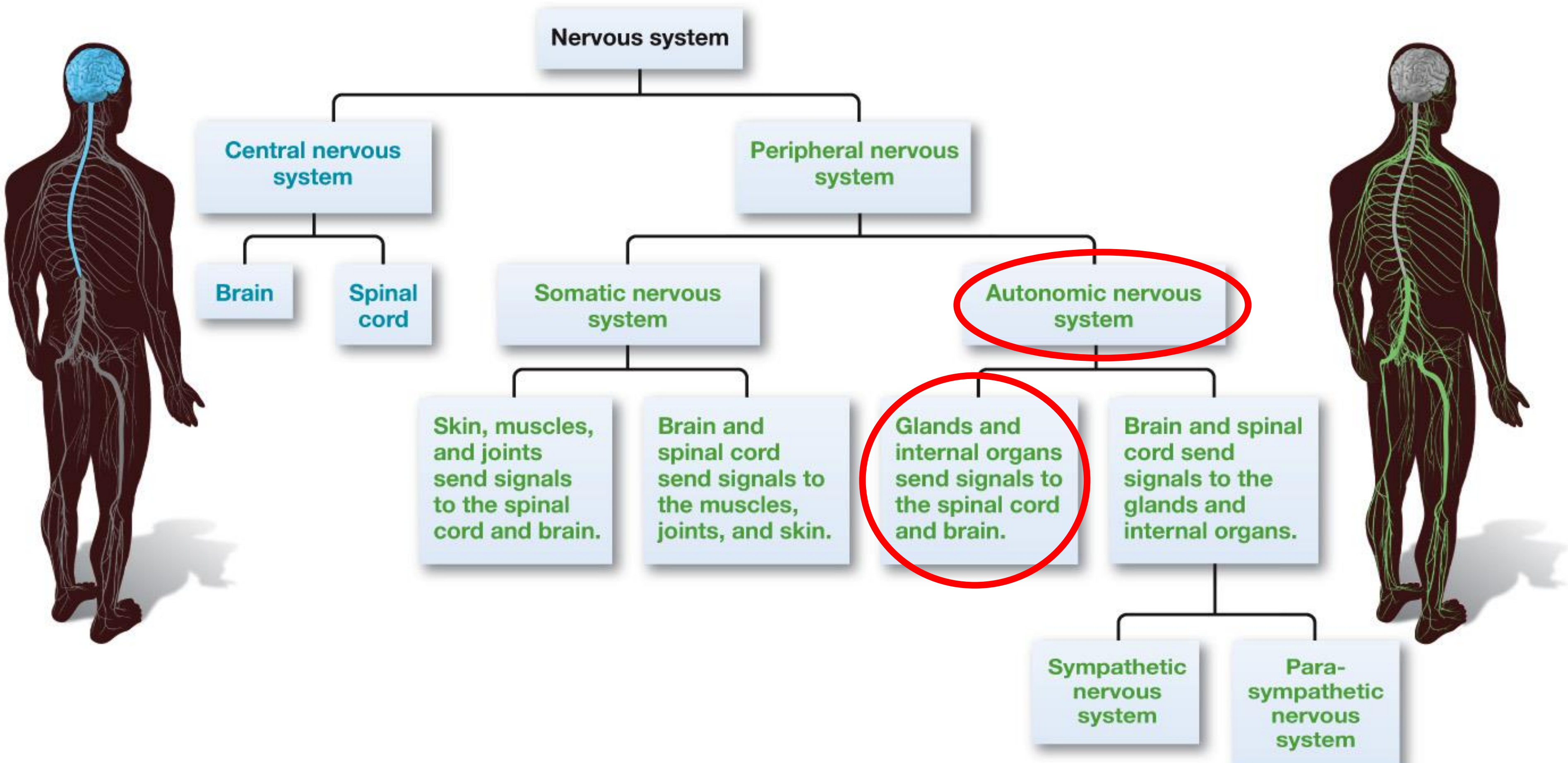


Communication of the brain with the body

- Somatic NS

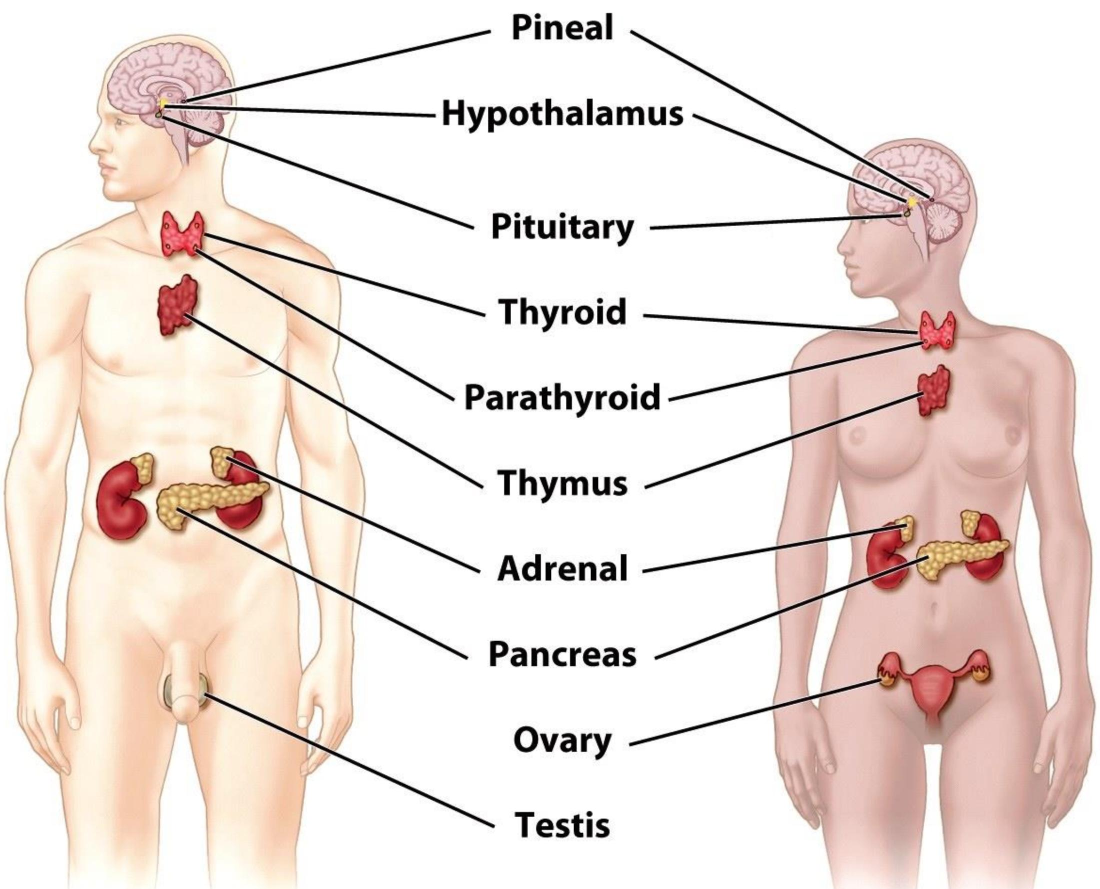


Communication of the brain with the body

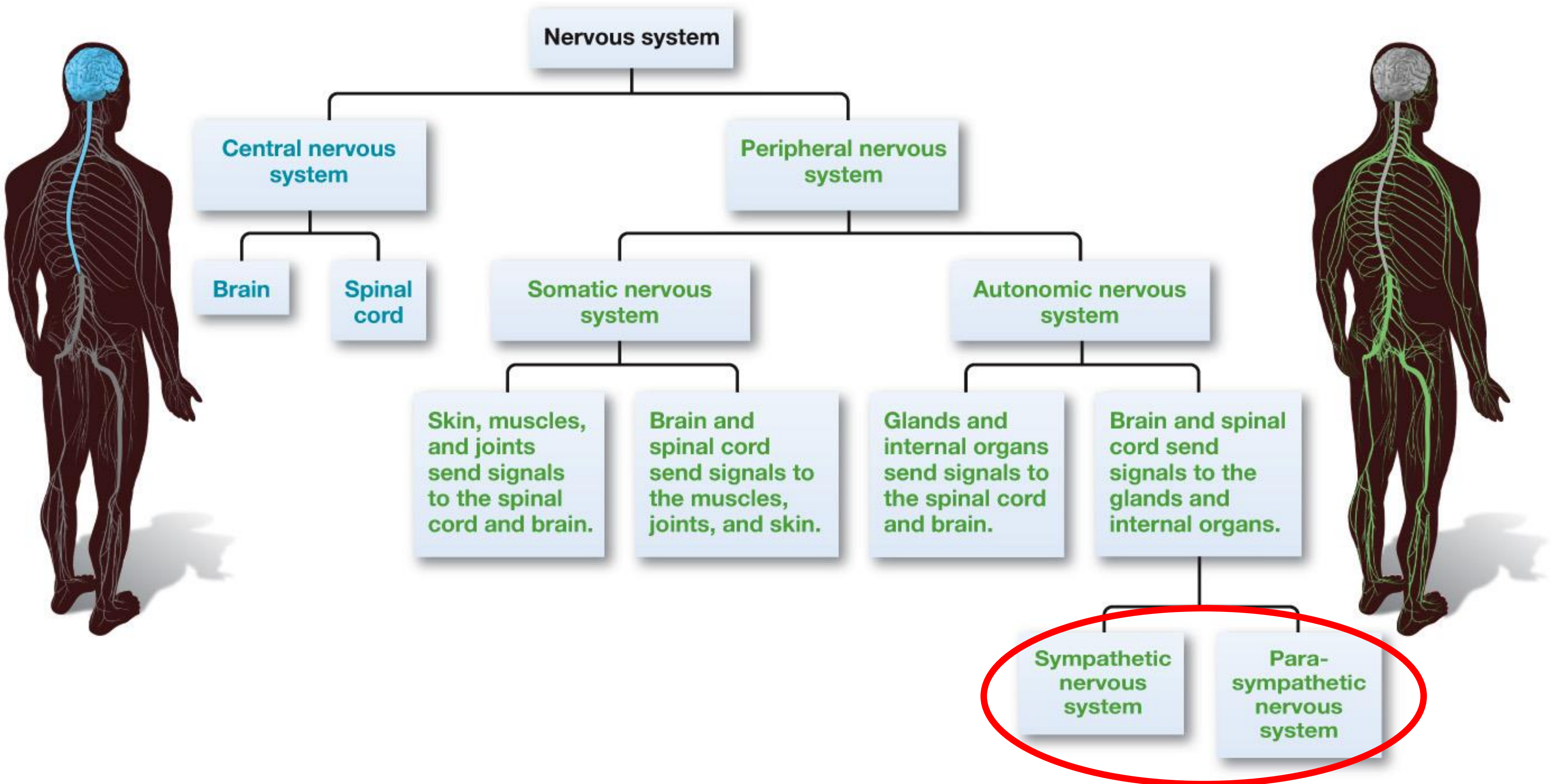


Communication of the brain with the body

- Endocrine system

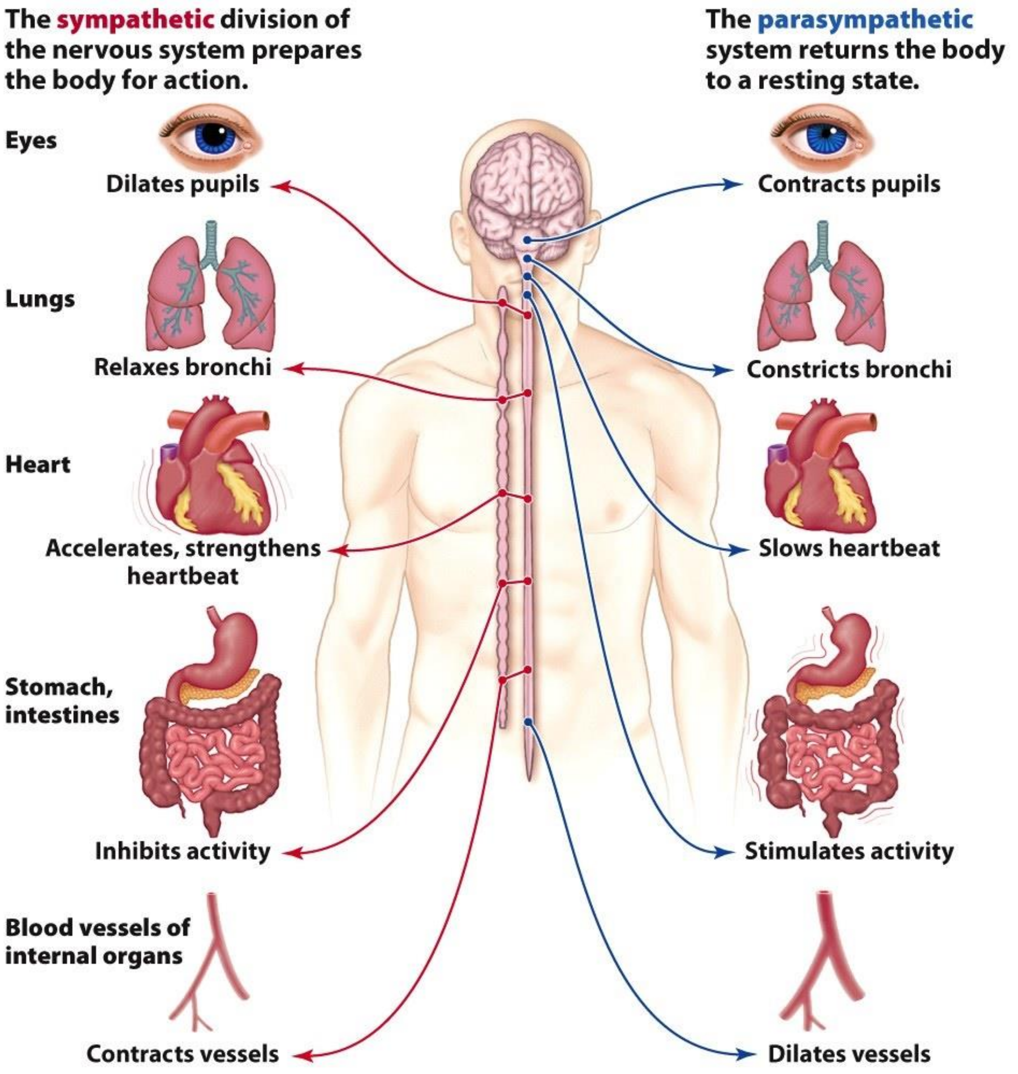


Communication of the brain with the body



Communication of the brain with the body

- Autonomic NS



Autonomic Nervous System (ANS)

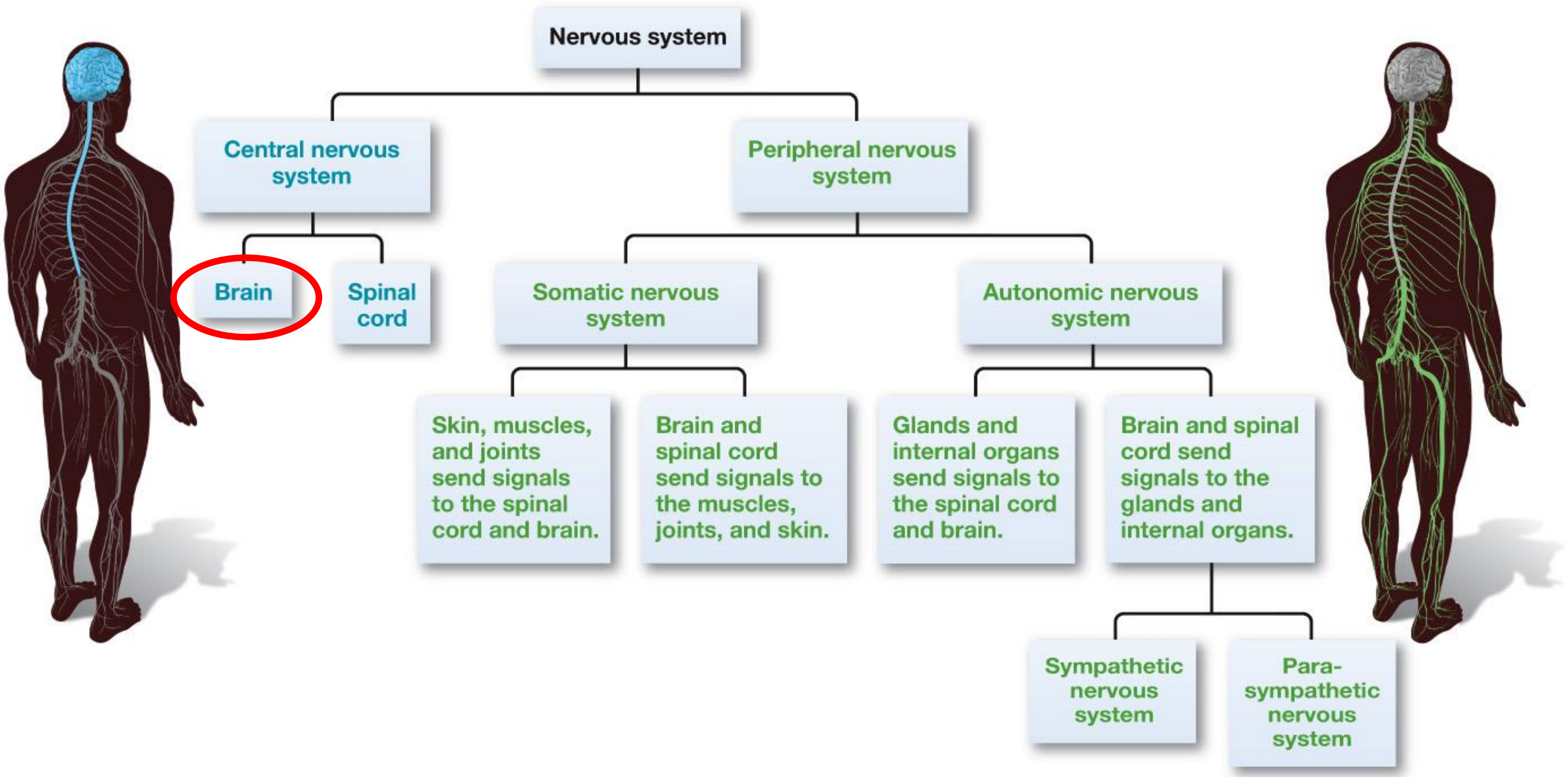
Sympathetic Nervous System:

- **Arouses** the body
  - mobilizing its energy in stressful situations
  - Uses energy
- In defensive situations
  - the heart rate increases
  - the lungs expand to hold more oxygen
  - the pupils dilate
  - blood flows to the muscles.

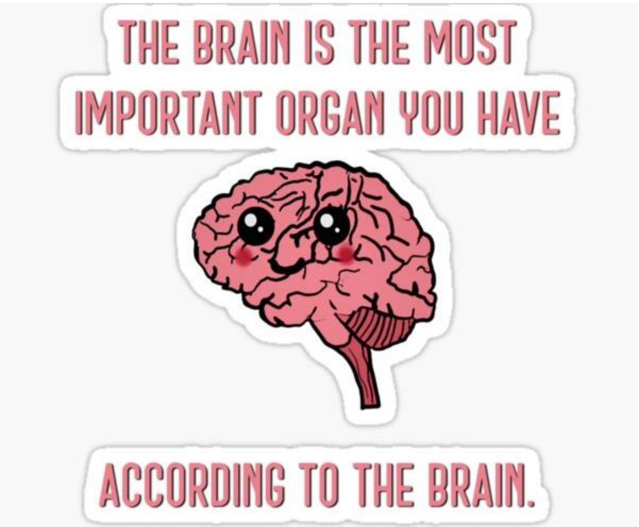
Parasympathetic Nervous System:

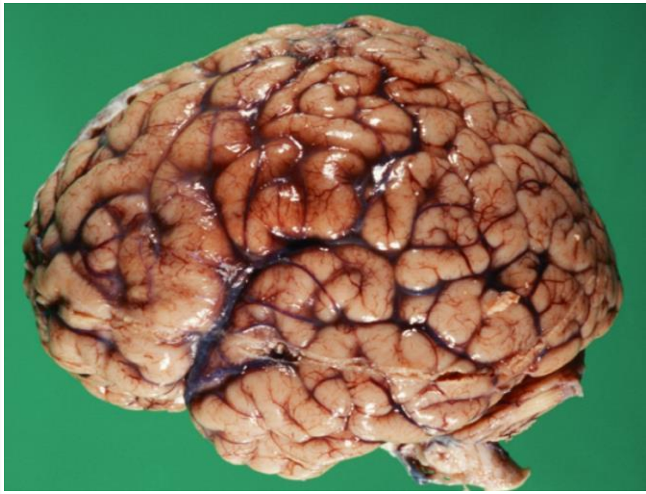
- **Calms** the body
  - conserving its energy
- Bodily functions that occur in normal, non-stressful situations.
- After eating
  - the digestive process begins:
  - nutrients are taken from the food and stored in the body

Studying the brain



The brain


THE BRAIN IS THE MOST
IMPORTANT ORGAN YOU HAVE
ACCORDING TO THE BRAIN.



The brain

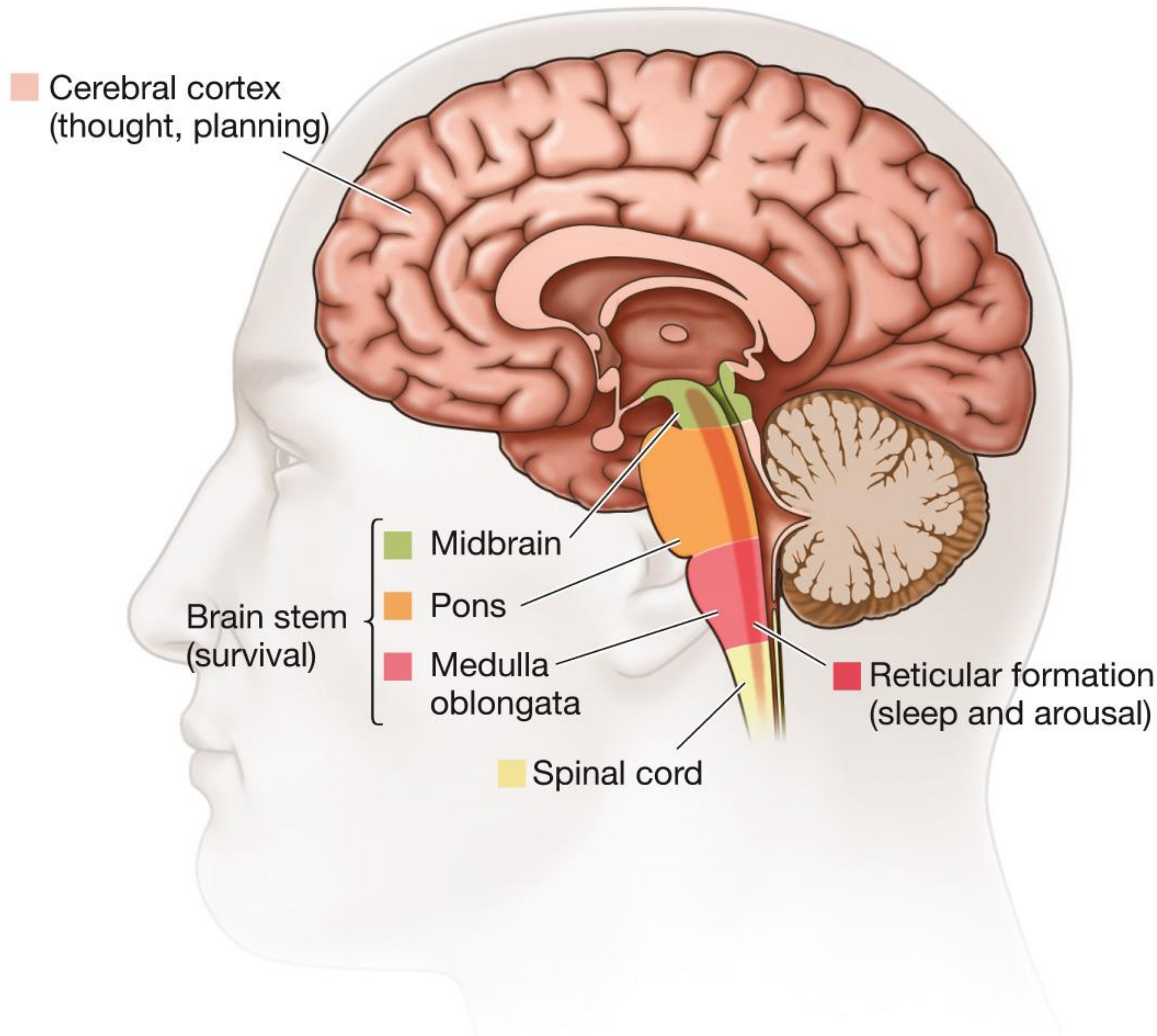
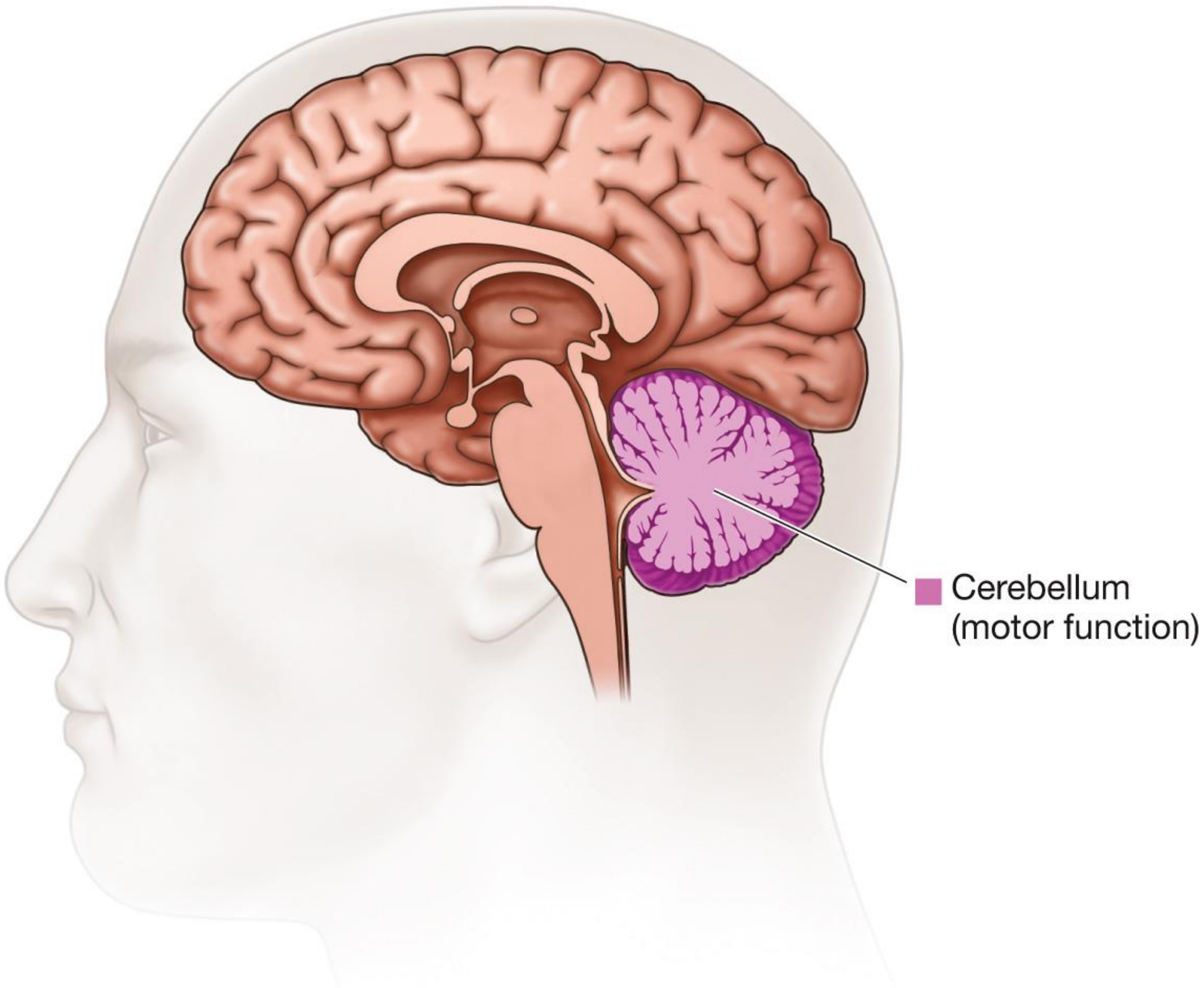
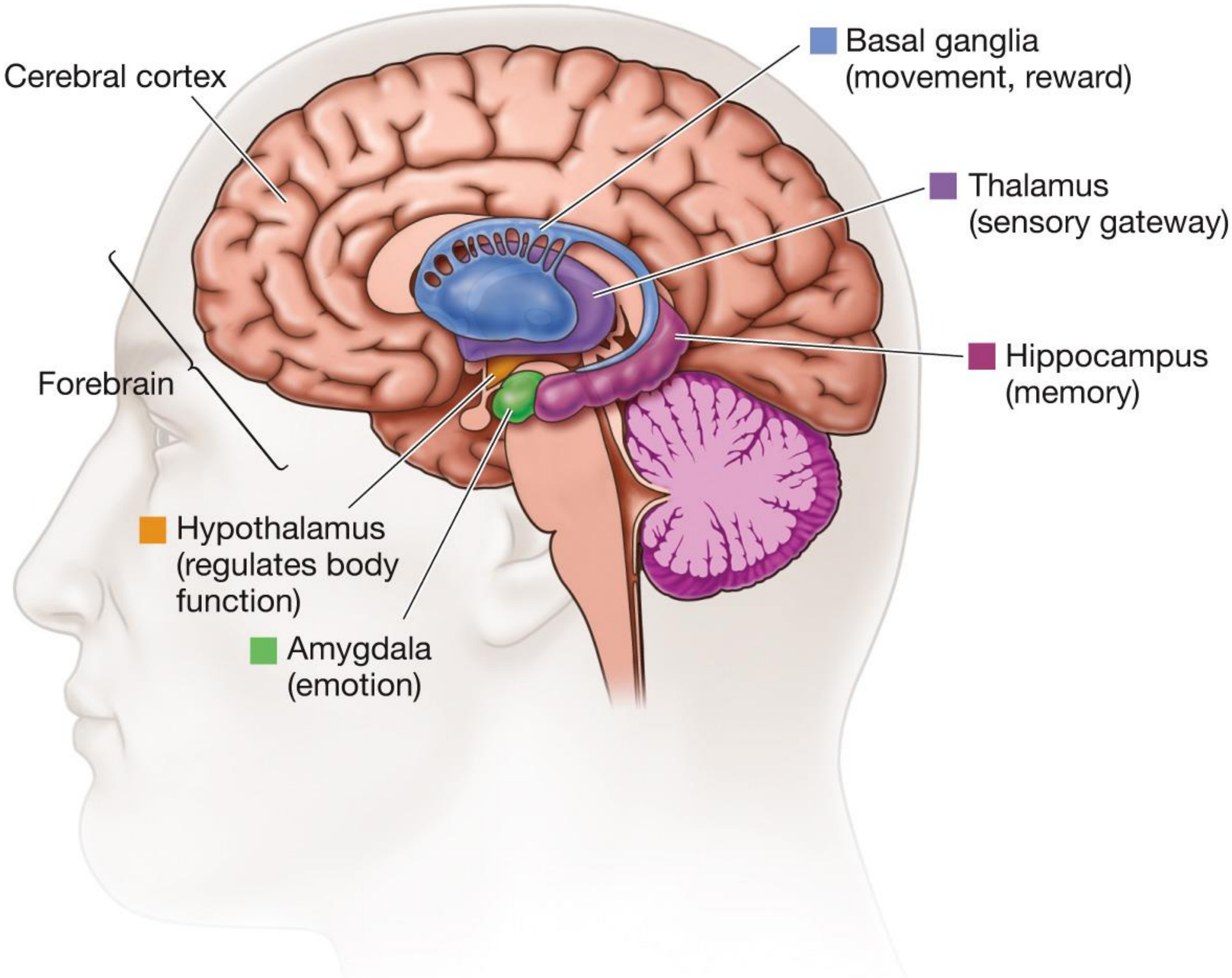


FIGURE 2.15 Myers/DeWall, *Psychology*, 12e,
© 2018 Worth Publishers
Andrew Swift

The brain

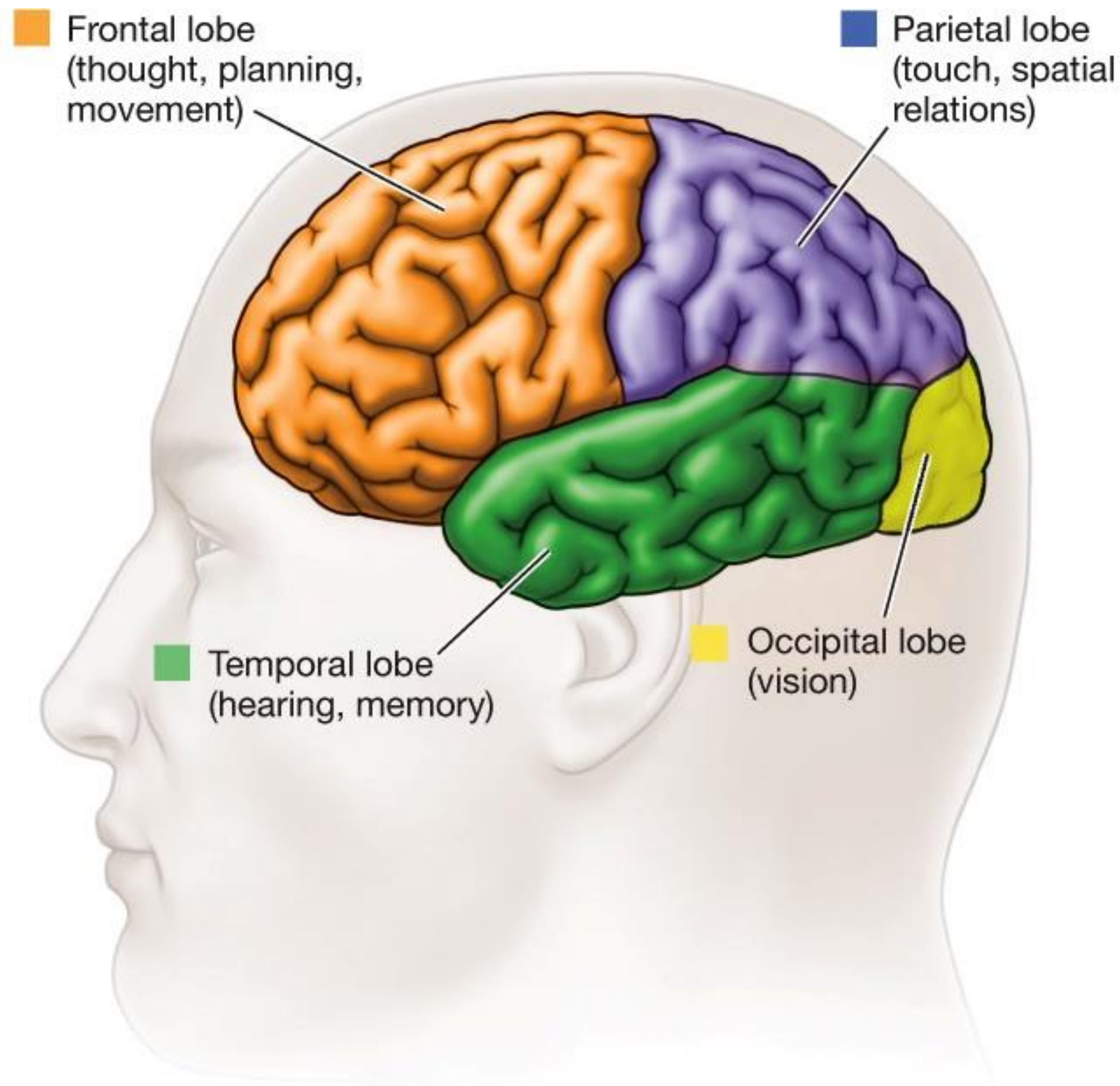


The brain

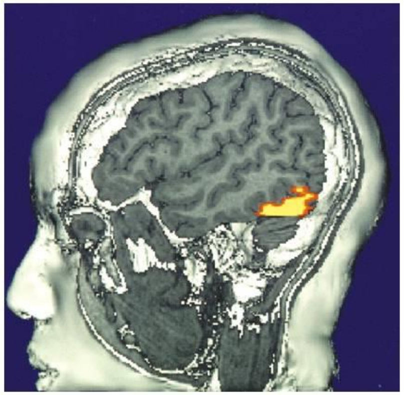


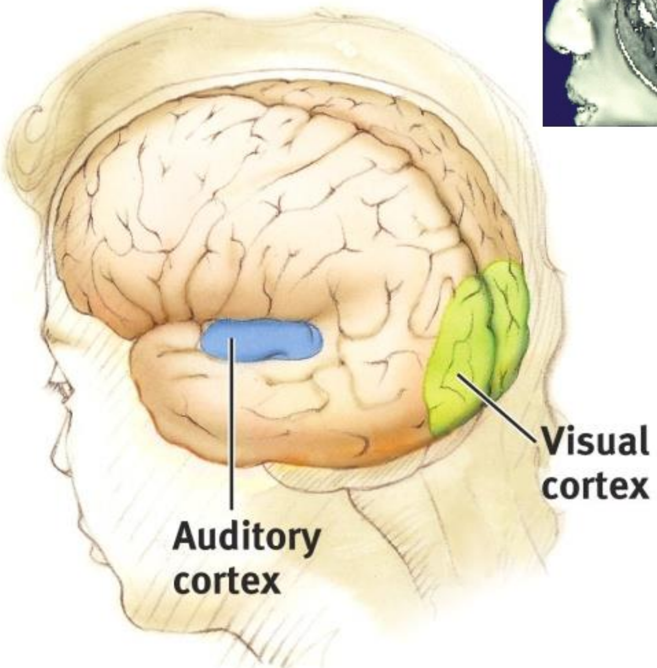
Cerebral cortex
Forebrain
Hypothalamus
(regulates body
function)
Amygdala
(emotion)
Basal ganglia
(movement, reward)
Thalamus
(sensory gateway)
Hippocampus
(memory)

The brain


Frontal lobe
(thought, planning,
movement)
Parietal lobe
(touch, spatial
relations)
Temporal lobe
(hearing, memory)
Occipital lobe
(vision)

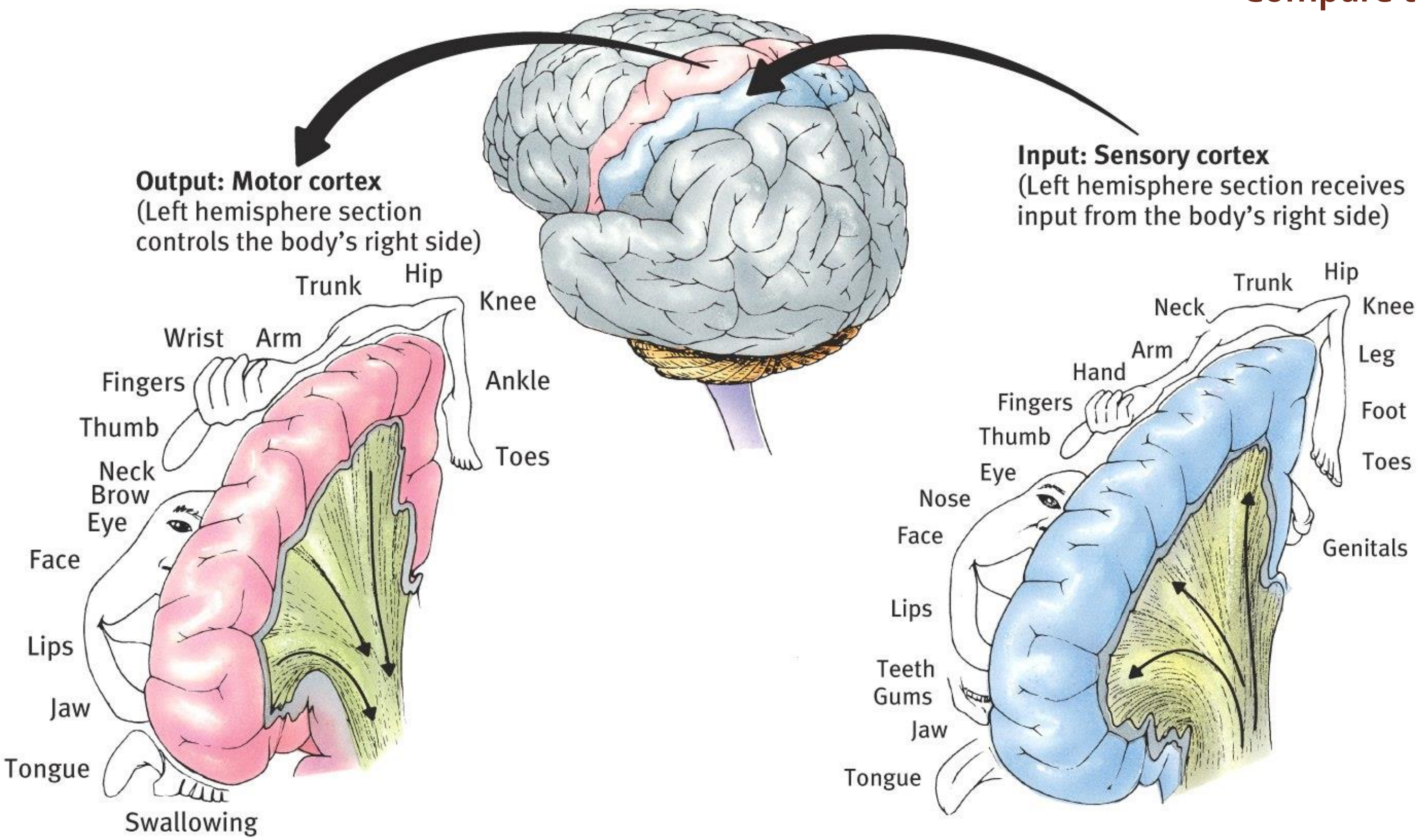
- **Visual Cortex**
  - The fMRI scan shows the visual cortex is active as the subject looks at faces.
  - Occipital lobe receives information from eyes and visual cortex gets activated

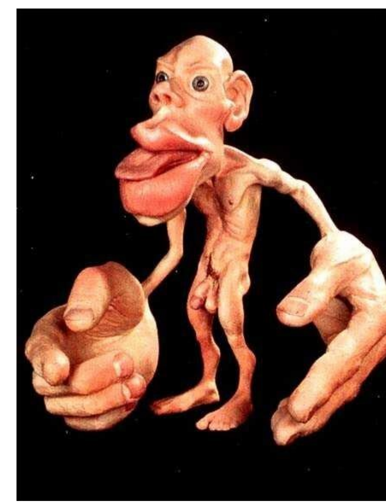




Motor Cortex & Sensory Cortex

- Larger sensory cortex is devoted to more sensitive body parts.
- Compare trunk and face.



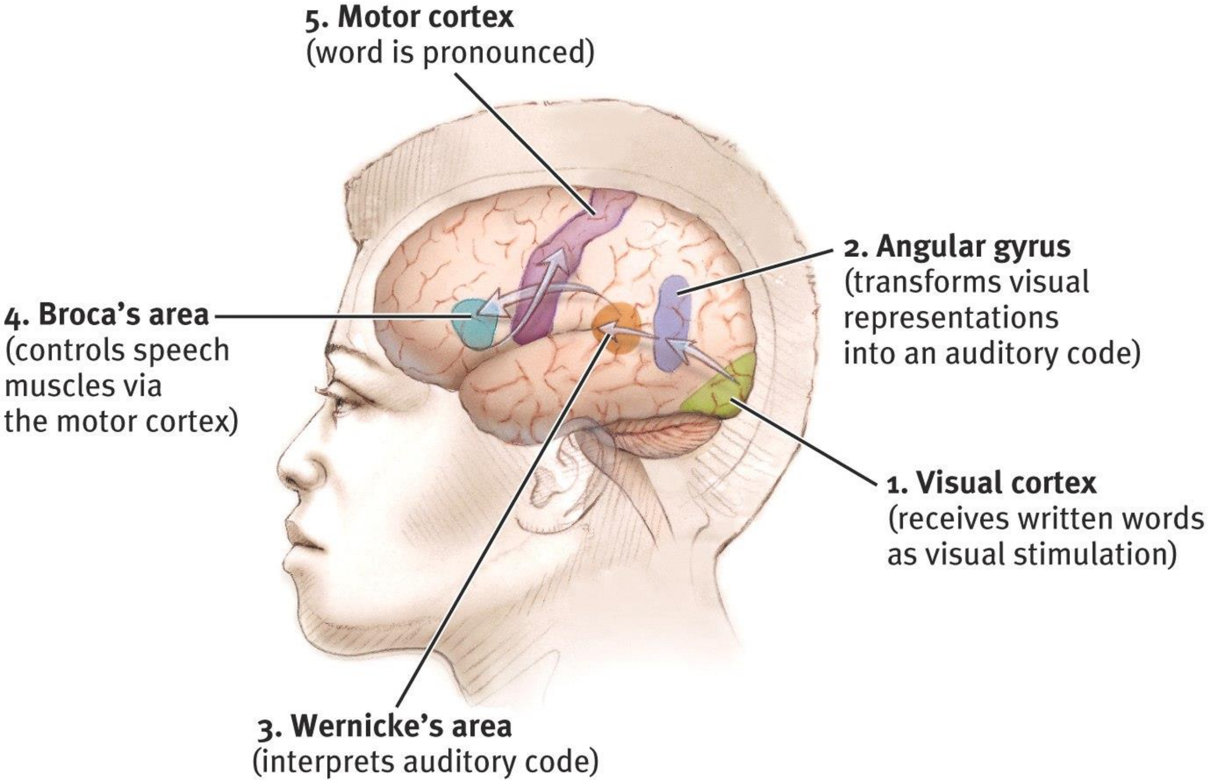


Cortical homonculus

Language

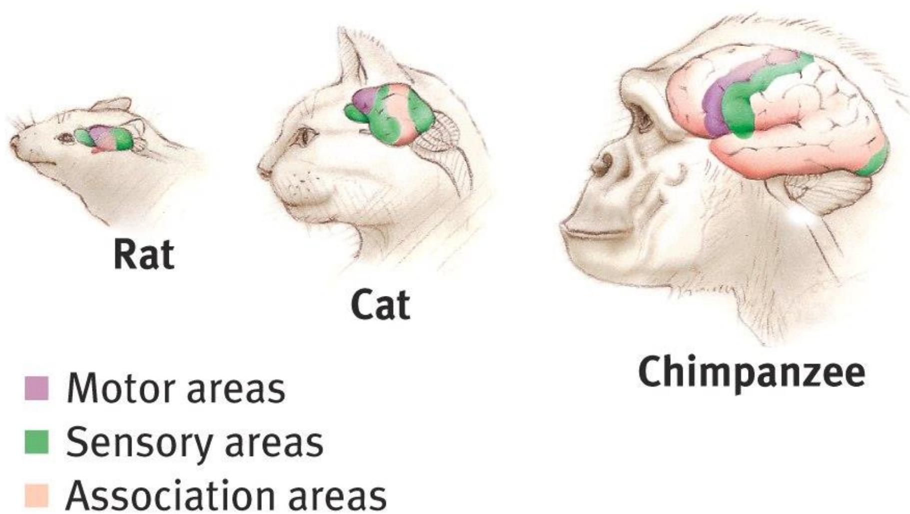
Broca's area:
impaired speaking

Wernicke's area:
impaired understanding



Association Areas

- More intelligent animals have increased association areas of the cortex: integrate and link inputs with stored memories.
- ¾ of the human cortex.
- Found in all lobes.



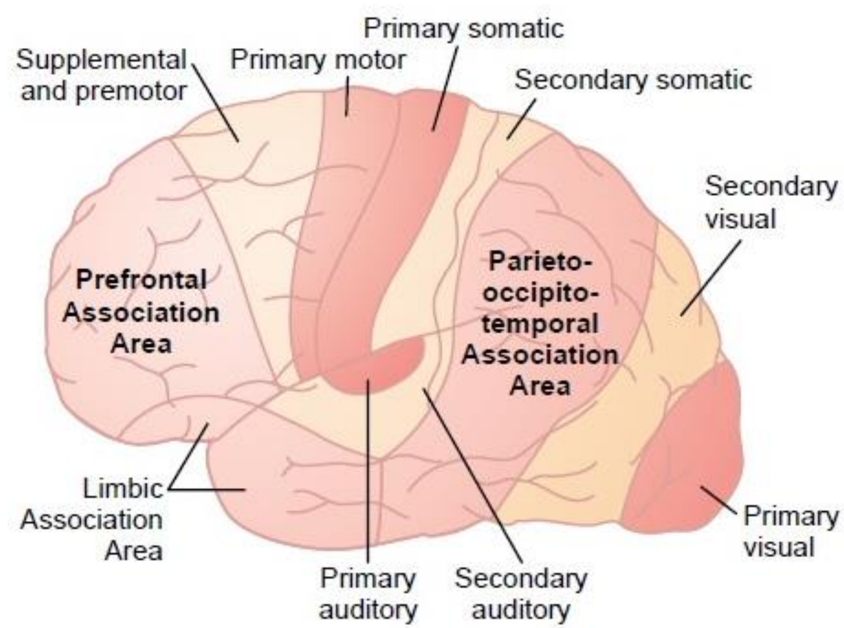
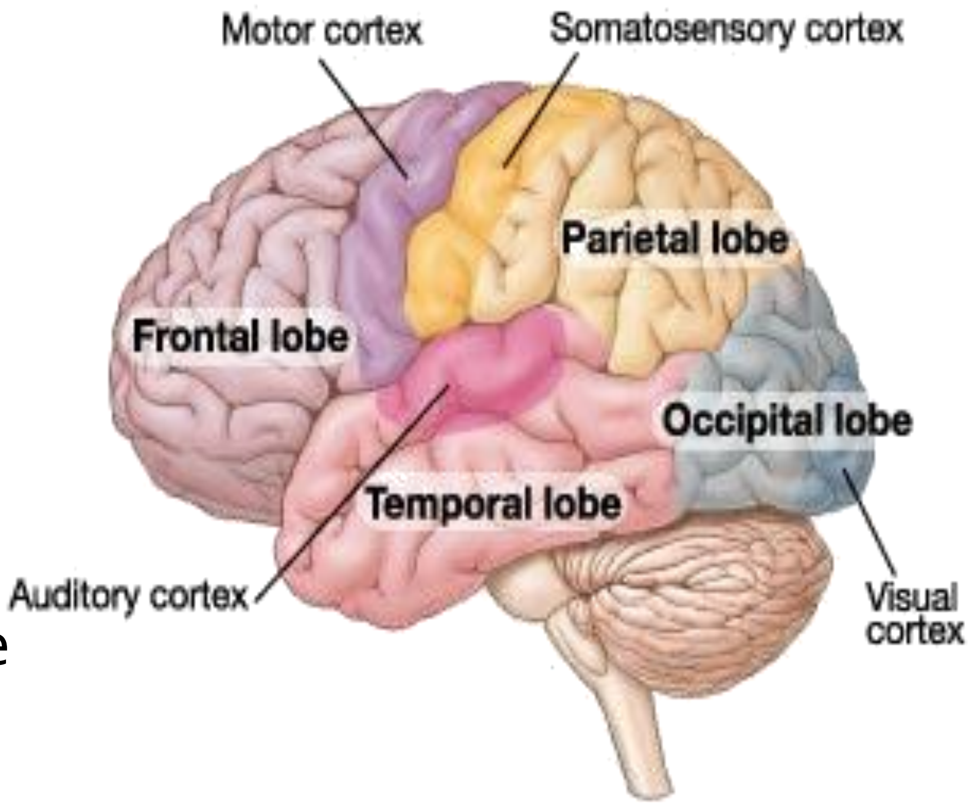


Figure 57–4

Locations of major association areas of the cerebral cortex, as well as primary and secondary motor and sensory areas.

Lobes of the Brain and Problems

- Frontal Lobe – personality, planning, emotion, problem solving
  - Motor cortex - movement
  - Broca's area – speech production
- Parietal Lobe - touch
- Temporal Lobe – hearing
  - Wernicke's area – language comprehension
- Occipital Lobe - vision



Brain Re-organization

- Plasticity
  - the brain’s capacity for modification, as evident in brain reorganization following

The message: “The brain is sculpted by our genes but also by our experiences.”

Our Divided Brain

Our brain is divided into two hemispheres.

The left hemisphere processes reading, writing, speaking, mathematics, and comprehension skills.

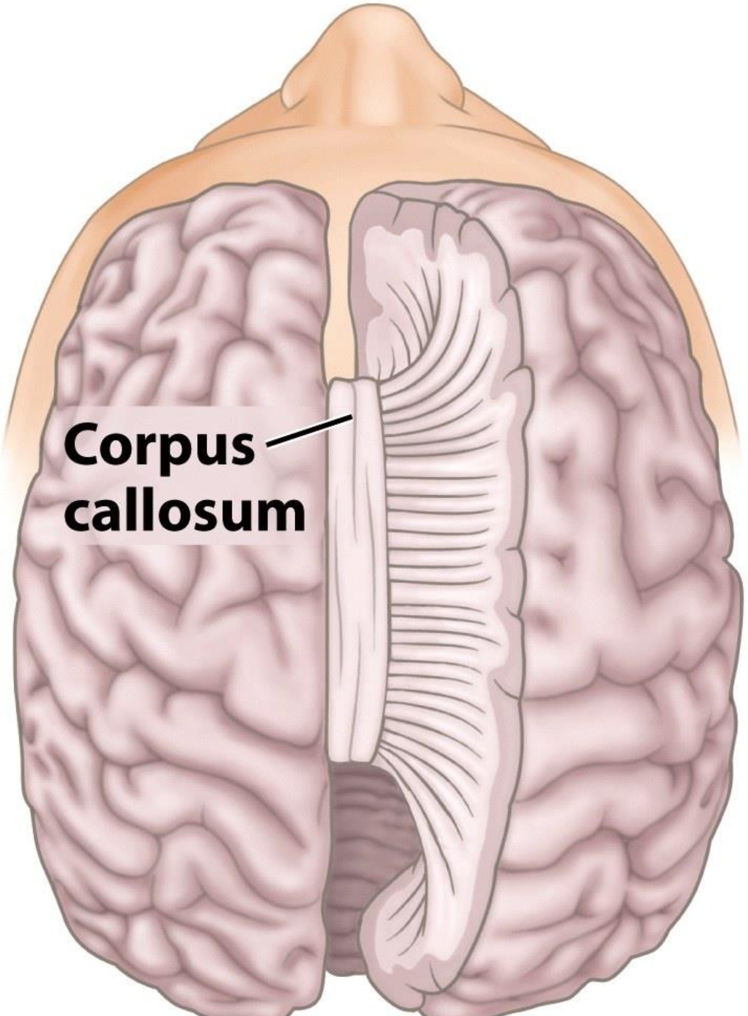
In the 1960s, it was termed as the <u>dominant brain.</u>

Hemispheric Specialization: Lateralization of Function

- **Left Hemisphere**
  - Language
  - Logic
  - Complex motor behavior.
- **Right Hemisphere**
  - Spatial ability: matching 3-D image with 2-D
  - Emotion: perceiving facial expressions and mood.
  - Musical ability: perception of melodies.
  - Some memory tasks: Learning tasks where context doesn’t matter.

Page 90 in your book

The brain

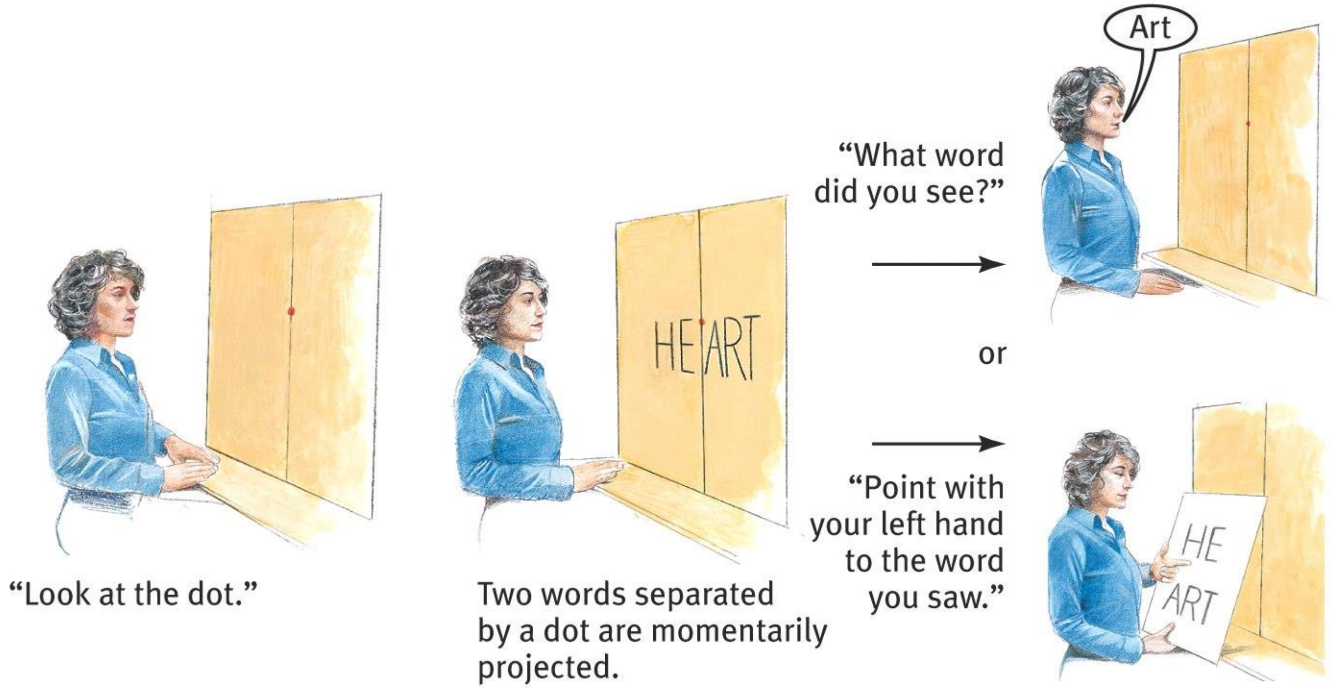


The brain

Split-Brain Experiment (Sperry & Gazzaniga)

- Subjects were presented information to one or the other side of their brains.
- Left hemisphere can tell what it has seen, right hemisphere can show it.
- Studies of split-brain patients:
  a. **Present a picture to the right visual field (left brain)**
     - Left hemisphere can tell you what it was.
     - Right hand can show you, left hand can't.
  b. **Present a picture to the left visual field (right brain)**
     - Subject will report that they do not know what it was.
     - Left hand can show you what it was, right can't.

Divided Consciousness



The brain

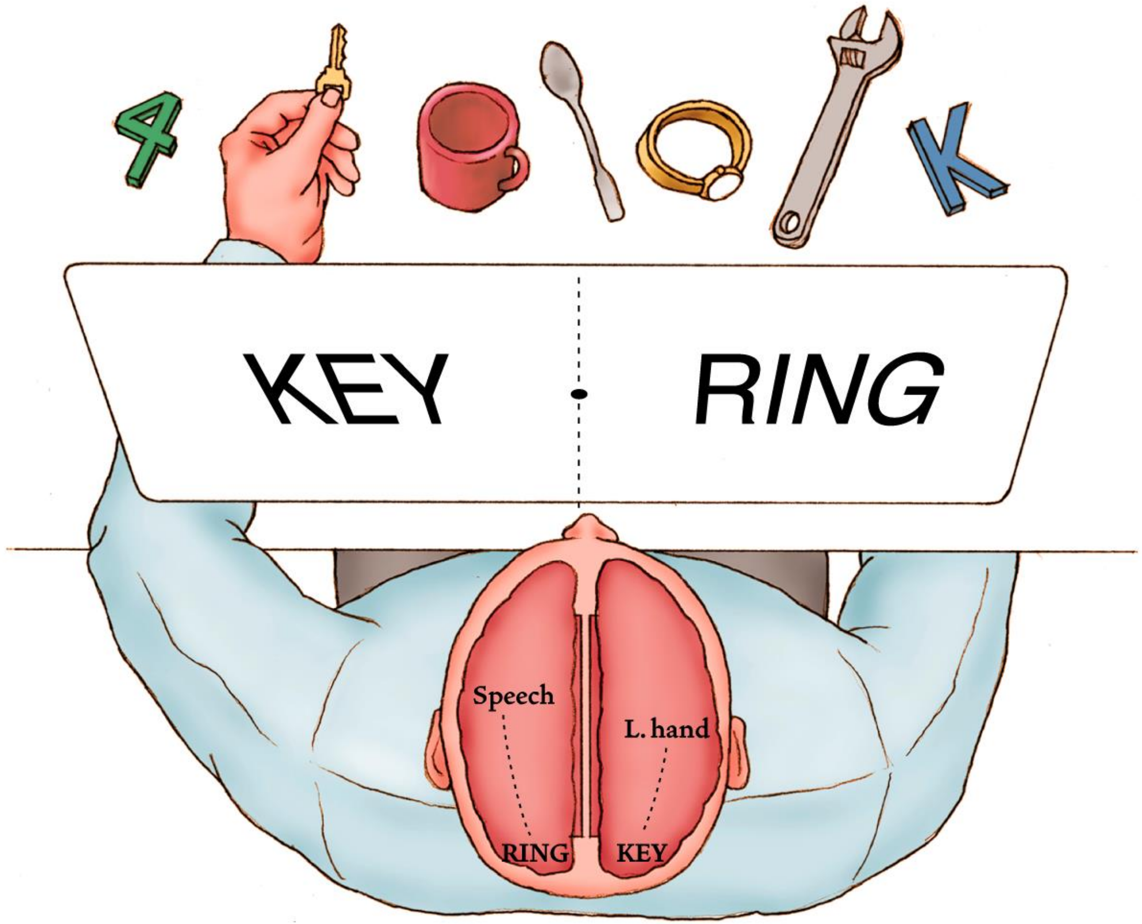


Figure 2.18: A setup sometimes used in split-brain studies

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SUMMARY

- Brain, the most complex structure of the human body, is the master of all neural processes took place in the body.

- Brain is capable of reorganizing itself in response to damage.

- Each hemisphere makes unique contributions to the integrated functioning of the brain.

When the instructor continues the lecture after the class is supposed to end